Motherson Automotive & Technology engineering
Survery No 172/1-5,274/2,276/7,
Chengalpattu Sriperumbudur Road, Vadgal Village B Vallam Panchayat,,Vadagal,
Kanchipuram, Tamil Nadu, 602105

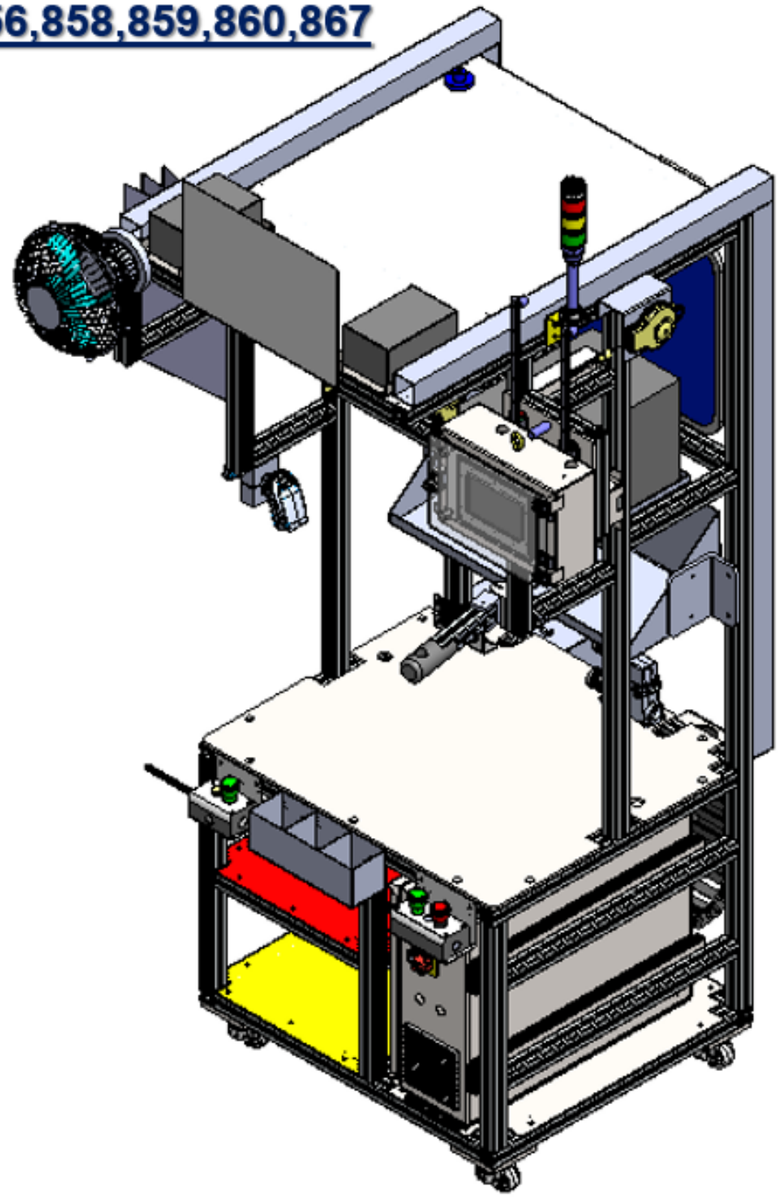
CUSTOMER NAME	MMLI PUNE
PROJECT NAME	S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1
PROJECT CODE	FMSPM00860
COMMISSION	ROBIS
PROJECT TYPE	WORK STATION
PLACE OF INSTALLATION	MMLI PUNE
RESPONSIBLE OF PROJECT	ROBIS
REVISION	01(Screw driver wiring updated)
DESIGNER	MANORAJ.S

CREATED ON	20-07-2023	CHECKED BY		
LAST EDITED DATE	11-04-2024	CHECKED ON	29-07-2023	NUMBER OF PAGES 32

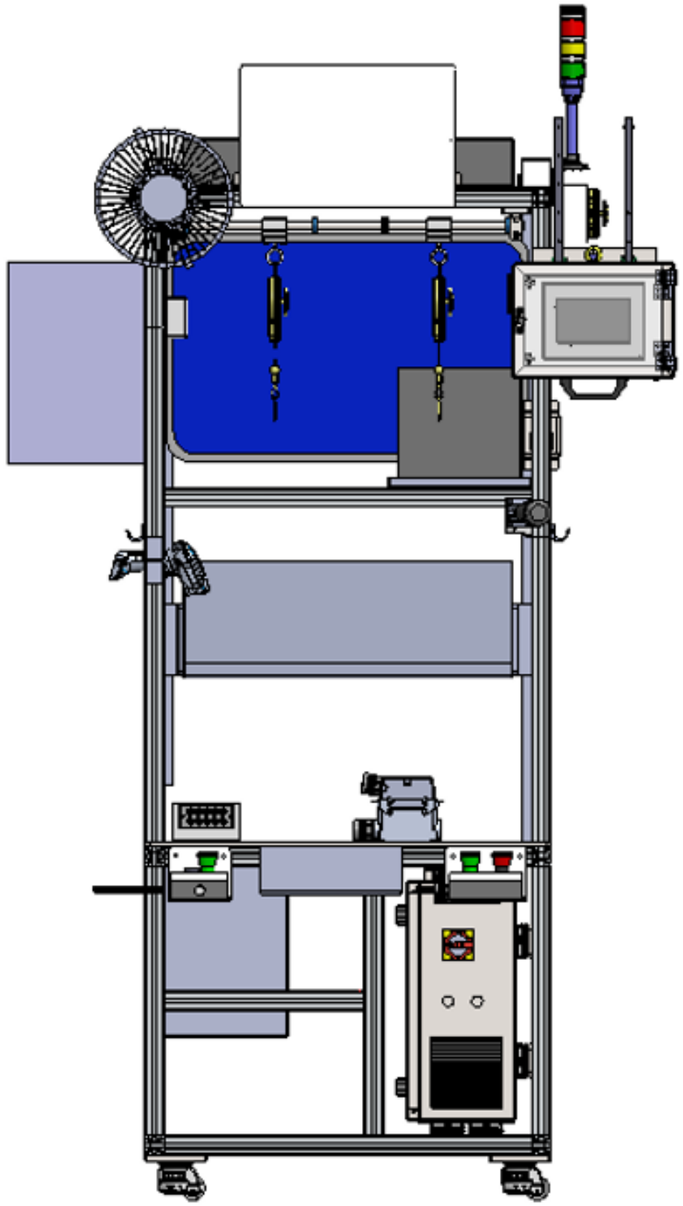
+CP/1

MACHINE OVER VIEW

FMSPM00856,858,859,860,867



ISO VIEW



FRONT VIEW

3

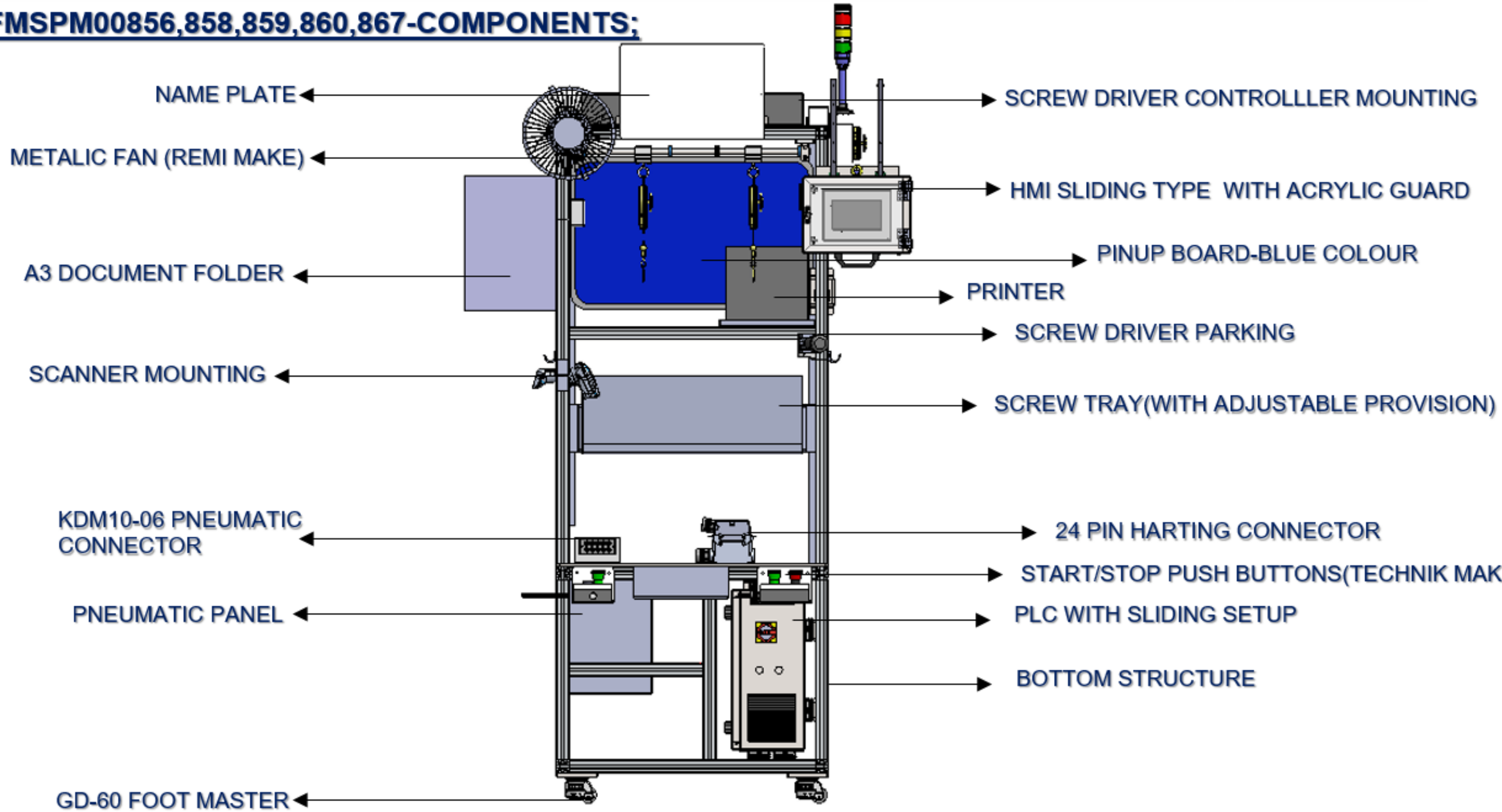
NOTE : PRINTER NOT CONSIDER

REVISION			Date	24-07-2023	ROBIS	Motherson Automotive & Technology engineering	MACHINE OVER VIEW		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 1
Modification	Date	Name	Original		Replacement of	Replaced by			Page 2 / 32

0	1	2	3	4	5	6	7	8	9

MACHINE FRONT VIEW

FMSPM00856,858,859,860,867-COMPONENTS;



NOTE : PRINTER NOT CONSIDER

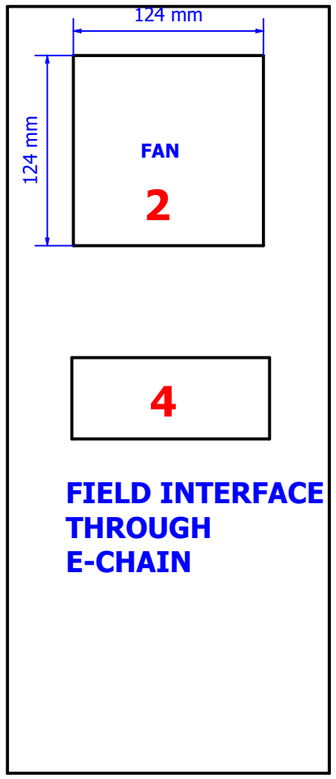
1 3

REVISION			Date	24-07-2023	ROBIS		Motherson Automotive & Technology engineering	MACHINE FRONT VIEW		=
			Ed.	MANORAJ.SELVARAJ						+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1				FMSPM00860	Page 2
Modification	Date	Name	Original		Replacement of	Replaced by				Page 3 / 32

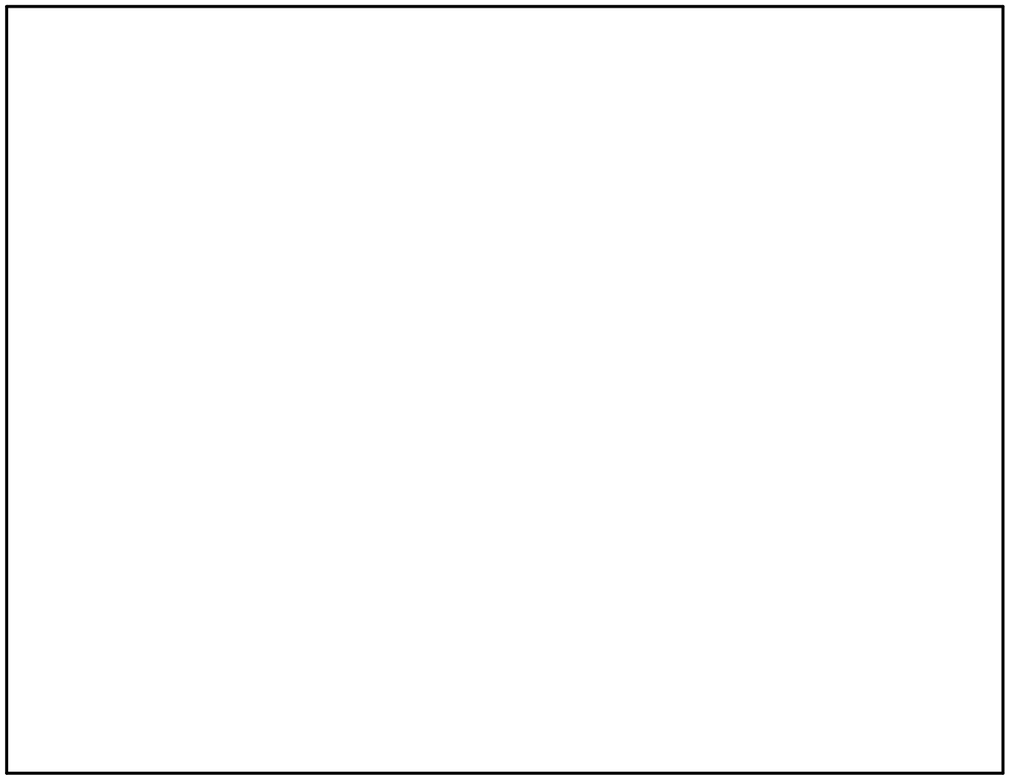
PANEL CUTOUT

CUSTOMIZED PANEL
H 500 X W 650 X D 210

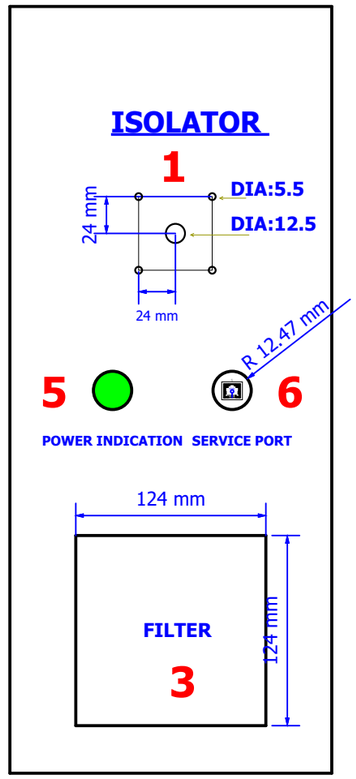
LEFT SIDE VIEW



FRONT SIDE VIEW



RIGHT SIDE VIEW

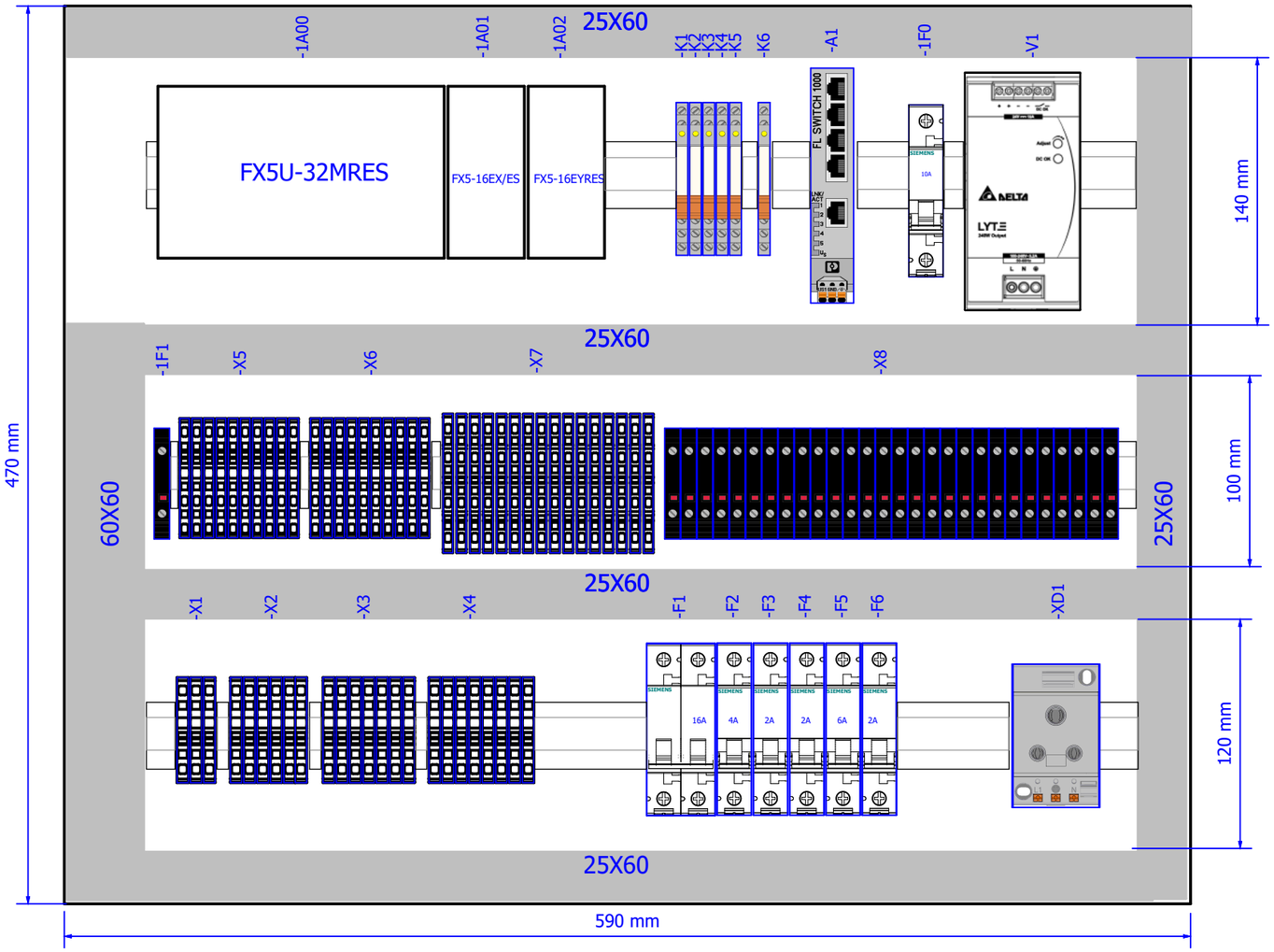


1	ISOLATOR CUTOUT
2	PANEL FAN CUTOUT
3	PANEL FILTER CUTOUT
4	FIELD INTERFACE CUTOUT
5	POWER INDICATION
6	SERVICE PORT

REVISION			Date	24-07-2023	ROBIS	Motherson Automotive & Technology engineering	PANEL CUTOUT		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 3
Modification	Date	Name	Original		Replacement of	Replaced by			Page 4 / 32

LAYOUT DIAGRAM

-1A00	CAAAZ00000000077 PLC FX5U32MRES MITSUBHISHI	1
-1A01	CAAAZ00000000072 16POINT INPUT CARD FX5-16EX/ES -MITSUBHISHI	1
-1A02	CAAAZ00000000073 16POINTS OUTCARD :-FX5-16EYRES, MISTUBHISHI	1
-K1 TO -K6	BMELES0935 SLIM RELAY COIL 24VDC	6
-A1	BMELES1310 ETEHRNET SWITCH 100 MB/S	1
-1F0	BMELES1708 MCB SINGLE POLE 10A SIEMENS MAKE	1
-V1	BMELES1532 DELTA LITE POWER SUPPLY 10A ,1 Phase ,240W	1
-1F1	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	1
-X1	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	2
-X1	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	1
-X2	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	6
-X3	BMELES0589 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT	7
-X4	BMELES0595 TERMINAL BLOCK 4.0 SQ MM 1 IN 1 OUT YE/GN	8
-X5	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	10
-X6	BMELES0364 TERMINAL BLOCK 2.5 SQ MM 1 IN 2 OUT	10
-X7	BMELES0366 TERMINAL BLOCK 2.5 SQ MM 2 IN 2 OUT	16
-X8	BMELES0584 3075799 TB4-HESILED 24 (5X20) EI	29
-F1	BMELES3072 SIEMENS MCB 2POLE 16AMPS 5SL62167RC	1
-F2	BMELES1706 MCB SINGLE POLE 4A SIEMENS MAKE	1
-F3	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F4	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
-F5	BMELES1707 MCB SINGLE POLE 6A SIEMENS MAKE	1
-F6	BMELES1705 MCB SINGLE POLE 2A SIEMENS MAKE	1
XD1	CAABA00000000433 804000 INDIAN 6A DIN RAIL POWER SOCKET	1



REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	LAYOUT DIAGRAM		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 4
Modification	Date	Name	Original		Replacement of	Replaced by			Page 5 / 32

PROJECT ARCHITECTURE



CAAAZ00000000077
PLC FX5U32MRES
MITSUBHISHI



CAAAZ00000000072
16POINT INPUT CARD
MODEL :-FX5-16EX/ES
MITSUBHISHI



CAAAZ00000000073
16POINTS OUTCARD
MODEL:-FX5-16EYRES,
MISTUBHISHI



BMELES1540
Mushroom Red Turn
Teknic 1NC 2PSML4 01
(2AML4+S2)



TOWERLIGHT 2
STG RG24V BUZ MNT
BMELES0523



BMELES1552
MUSHROOM PB GREEN
Teknic 1NO 2PSM3 10
(2AM3+S1)



BMELES1552
MUSHROOM PB GREEN
Teknic 1NO 2PSM3 10
(2AM3+S1)



BMELES1542
NONIII. YELLOW PUSH
BUTTON, Teknic 1 NO
2PSF8 10 (2AF8+S1)



SENSORS
(CUSTOMER
SCOPE)



BMPNES5128
NEBU-M8G3-K-5-LE3



SCREW DRIVER 1
(CUSTOMER
SCOPE)



BMELES1310
Ethernet Switch
100 MB/S



CAAAI00000000586
HMI MODEL : GS2107-WTBD
MITSUBHISH



BMELES3959
Data logic power
scanner PD9531

INPUT/OUTPUT
SAFETY
ETHERNET COMMUNICATION
RS 232 COMMUNICATION

4 6

REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	PROJECT ARCHITECTURE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 5
Modification	Date	Name	Original		Replacement of	Replaced by			Page 6 / 32

MAIN POWER DISTRIBUTION



-F1
BMELES3072
SIEMENS MCB 2POLE 16AMPS
5SL62167RC



-Q1
CAABA00000000274
16 A /3P /1WAY ISOLATOR
ROTARY SWITCH

-X1
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-W0
3Gx2.5 mm²
BMELES1940

BMELES3408
INDUSTRIAL
SOCKET LEGRAND

INDUSTRIAL SOCKET

POWER SUPPLY
1 PHASE 230V
50-60 Hz

-X2
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-X3
BMELES0589
TERMINAL BLOCK CX4
4.0 SQ MM 1 IN 1 OUT

-X4
BMELES0595
TERMINAL BLOCK
4.0 SQ MM 1 IN 1 OUT YE/GN

HMI PANEL EARTH

PNEUMATIC PANEL EARTH

PANEL EARTH

5 7

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN POWER DISTRIBUTION	=	
			Ed.	manoraj.selvaraj				+ CP	
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 6
Modification	Date	Name	Original		Replacement of	Replaced by			Page 7 / 32

POWER SUPPLY

-V1
BMELES1532
DELTA LITE POWER SUPPLY
10A ,1 Phase ,240W



Weidmüller PROeco
L(+) N(-) PE
Adjust 22-28 V
+ + - -
13 14

1L1-1 / 6.2 -X2:1:B
1N1-1 / 6.5 -X3:1:B
PE-1 / 6.7 -X4:1:B

1L1-1 BK 2.5 mm²
1N1-1 BU 2.5 mm²
PE-1 GNYE 2.5 mm²

-F2
MCB SINGLE POLE
4 A SIEMENS MAKE
5SL6104RC;
BMELES1706

2L1-1 BK 2.5 mm²
1N1-1 BU 2.5 mm²
PE-1 GNYE 2.5 mm²

2L1-1 1.5 mm² DBU 24VDC
1N1-1 1.5 mm² DBU P0VDC
PE-1 2.5 mm² GNYE PE-3

-1F0:2 10.2 / 24VDC
-X6:1 10.6 / P0VDC
-X4:3:B 6.8 / PE-3

1L1-2 / 6.3 -X2:2:B
1N1-2 / 6.5 -X3:2:B
PE-2 / 6.7 -X4:2:B

1L1-2 BK 2.5 mm²
1N1-2 BU 2.5 mm²
PE-2 GNYE 2.5 mm²

-F3
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE

2L1-2 BK/1 0.75 mm²
1N1-2 BK/2 0.75 mm²
PE-2 GNYE 0.75 mm²

-W1
BMELES0075
3 G 0.75 mm²

3L1-2 BK/1 0.75 mm²

-LS
BMELES1669
AE PANEL DOOR
LIMIT SWITCH -
SZ 4127010



21
22

-W2
BMELES0075
3 G 0.75 mm²

4L1-2 BK/2 0.75 mm²
1N1-3 BU 1 mm²

-M1
M
1~

BMELES0049
SK 3322107-
FAN AND FILTER UNIT



-H1
BMELES2527
1Feet 6W LED
TUBE LIGHT WITH
ALUMINIUM BATTERY (CW)

1N1-3 / 6.5 -X3:3:B
1L1-3 / 6.3 -X2:3:B

1L1-3 BK 2.5 mm²
1N1-3 BU 1.5 mm²

-F4
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE

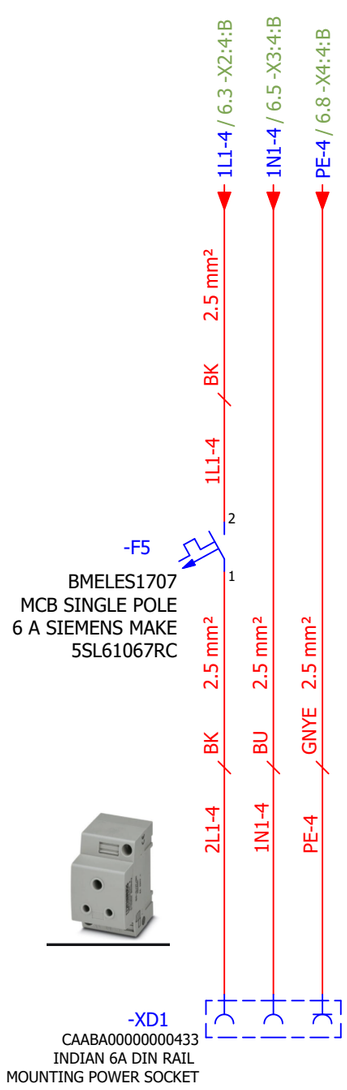
2L1-3 BK 1.5 mm²

-K6:11 21.1 / 2L1-3

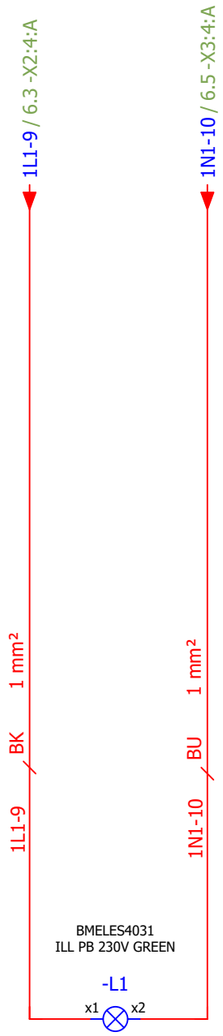
6 8

REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	POWER SUPPLY		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 7
Modification	Date	Name	Original		Replacement of	Replaced by			Page 8 / 32

INDICATION LAMP



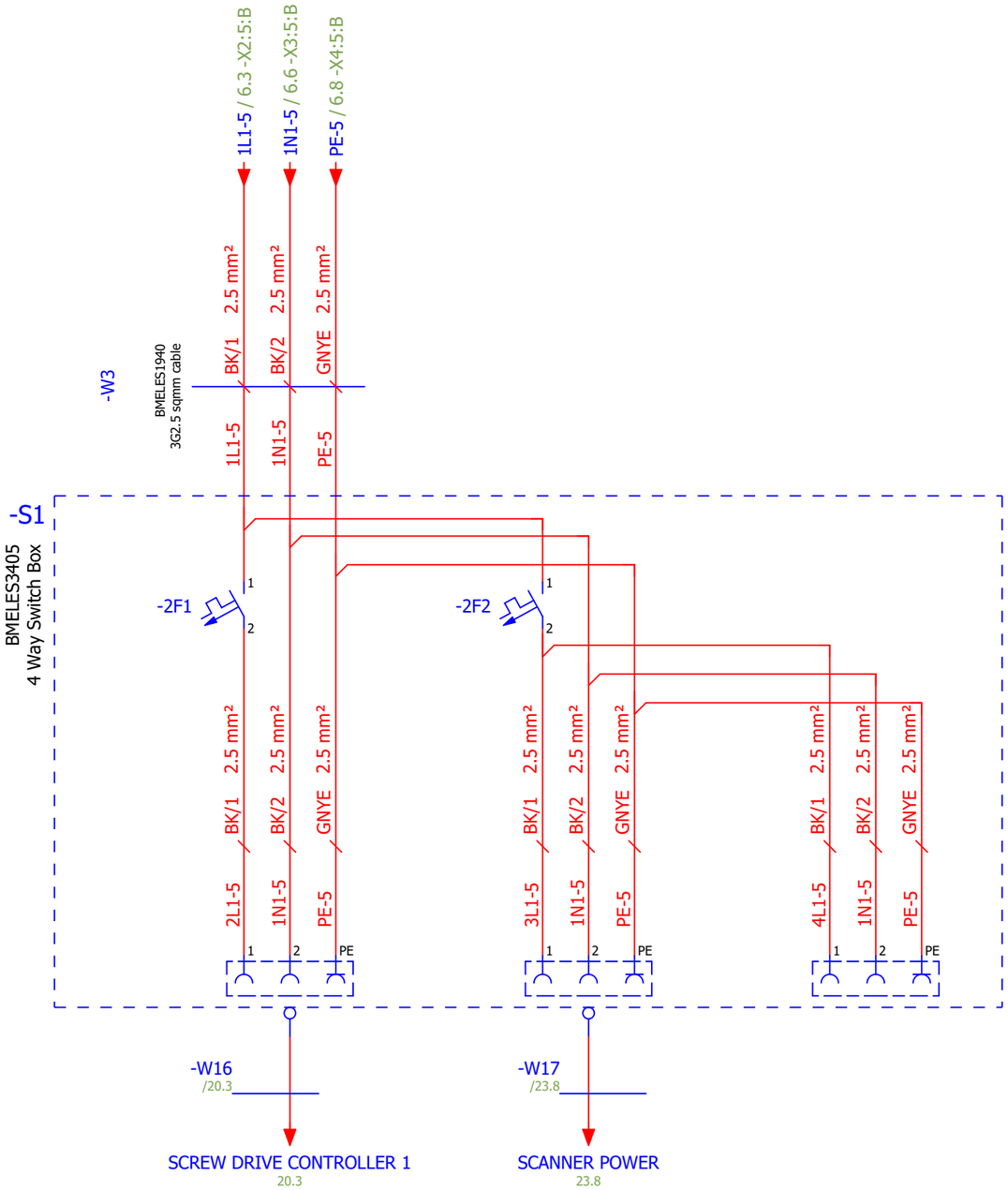
POWER SOCKET

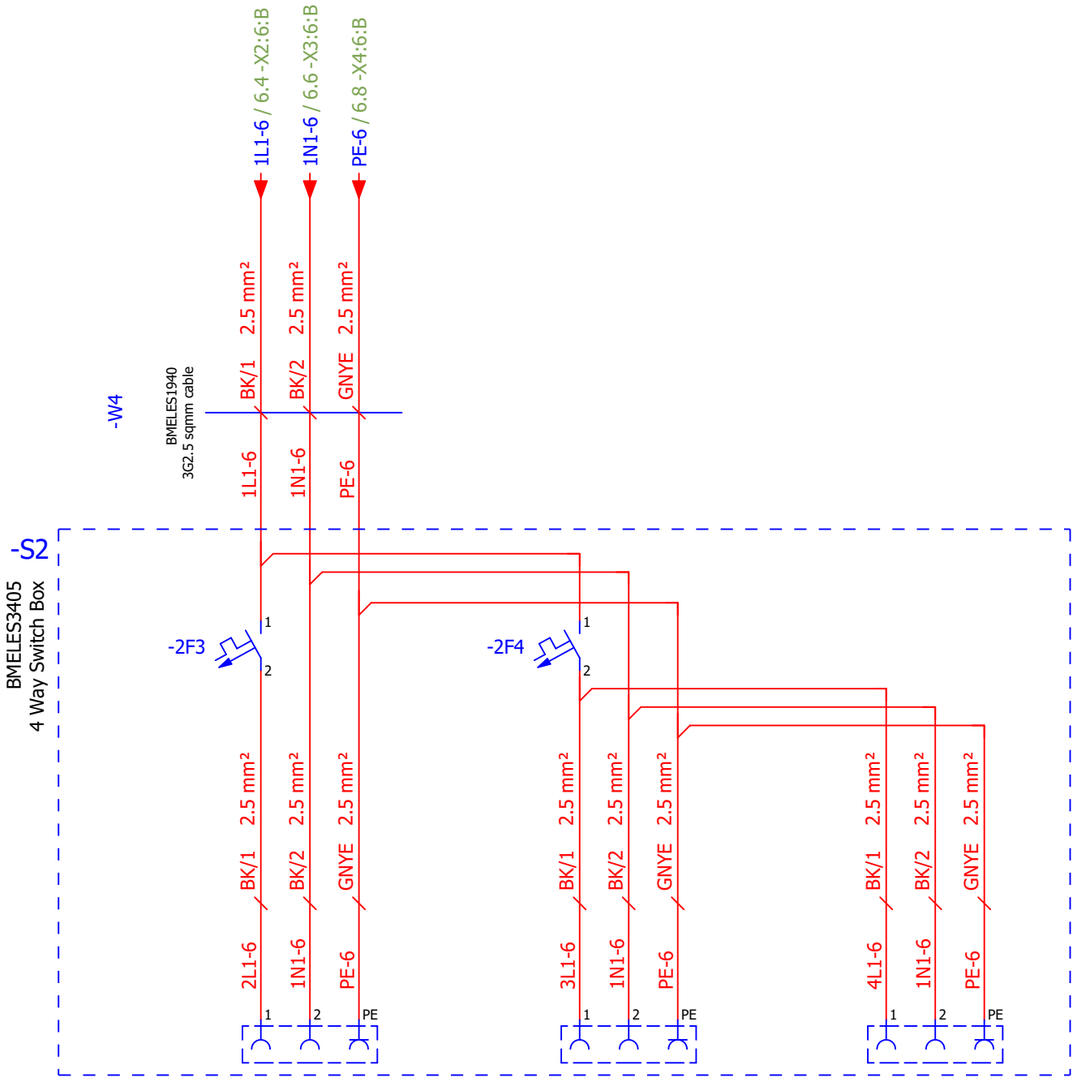


230V GREEN
POWER INDICATION
LAMP

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	INDICATION LAMP		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 8
Modification	Date	Name	Original		Replacement of	Replaced by			Page 9 / 32

SOCKET INTERFACE





8 10

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	SOCKET INTERFACE		=	
			Ed.	manoraj.selvaraj					+ CP	
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page	9
Modification	Date	Name	Original		Replacement of	Replaced by			Page	10 / 32

DISTRIBUTION 24VDC TERMINAL BLOCK

P24VDC1 / 10.2 -X5:1

P24VDC1 DBU 1.5 mm²

-1F1
2A
BMELES0584
TB4-HESILED
24 (5X20) EI

1P24VDC BK/1 0.75 mm²

-HMI:+ 22.1 / 1P24VDC



24VDC / 7.1 -V1:+

24VDC DBU 1.5 mm²

-1F0
BMELES1708
MCB SINGLE POLE 10A
SIEMENS MAKE
5SL61107RC

P24VDC DBU 1.5 mm²

-X5
BMELES0364
TERMINAL BLOCK
2.5 SQ MM 1 IN 2 OUT

P24VDC1	DBU	1.5 mm²	-1F1 10.0 / P24VDC1
P24VDC2	DBU	0.5 mm²	-1A00:COM0 14.1 / P24VDC2
P24VDC3	DBU	0.5 mm²	-1A00:COM2 14.5 / P24VDC3
P24VDC4	DBU	0.5 mm²	-1A02:+V4 16.1 / P24VDC4
P24VDC5	DBU	0.5 mm²	-1A02:+V6 16.5 / P24VDC5
P24VDC6	DBU	0.5 mm²	-K3:11 19.5 / P24VDC6
P24VDC7	DBU	0.5 mm²	-A1:US:1 11.1 / P24VDC7
P24VDC8	BK/1	0.75 mm²	-PB1-S2:13 24.4 / P24VDC8
P24VDC9	BK/2	0.75 mm²	-PB1-S1:11 24.5 / P24VDC9
P24VDC10	BK/1	0.75 mm²	-PB2-S4:13 24.1 / P24VDC10
P24VDC11	BK/2	0.75 mm²	-PB2-S3:13 24.2 / P24VDC11
P24VDC12	BK/1	0.75 mm²	-PSC 22.5 / P24VDC12
P24VDC13	BK/23	1 mm²	-1X100:16 25.5 / P24VDC13

P0VDC / 7.1 -V1:-

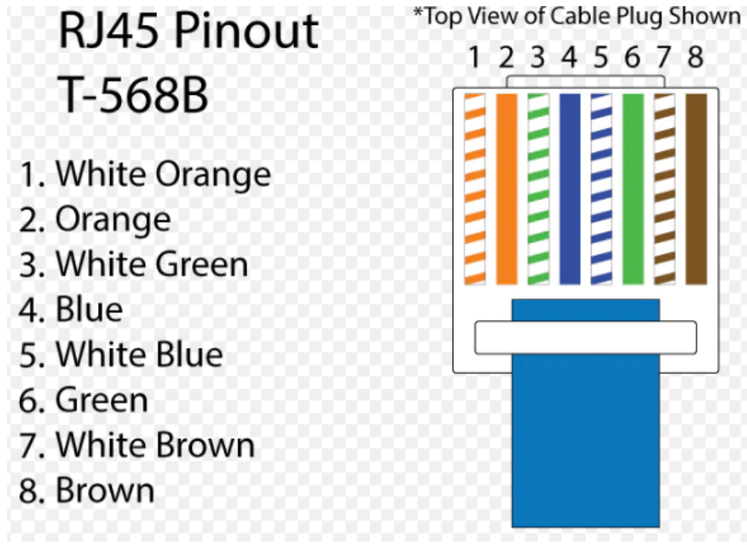
P0VDC DBU 1.5 mm²

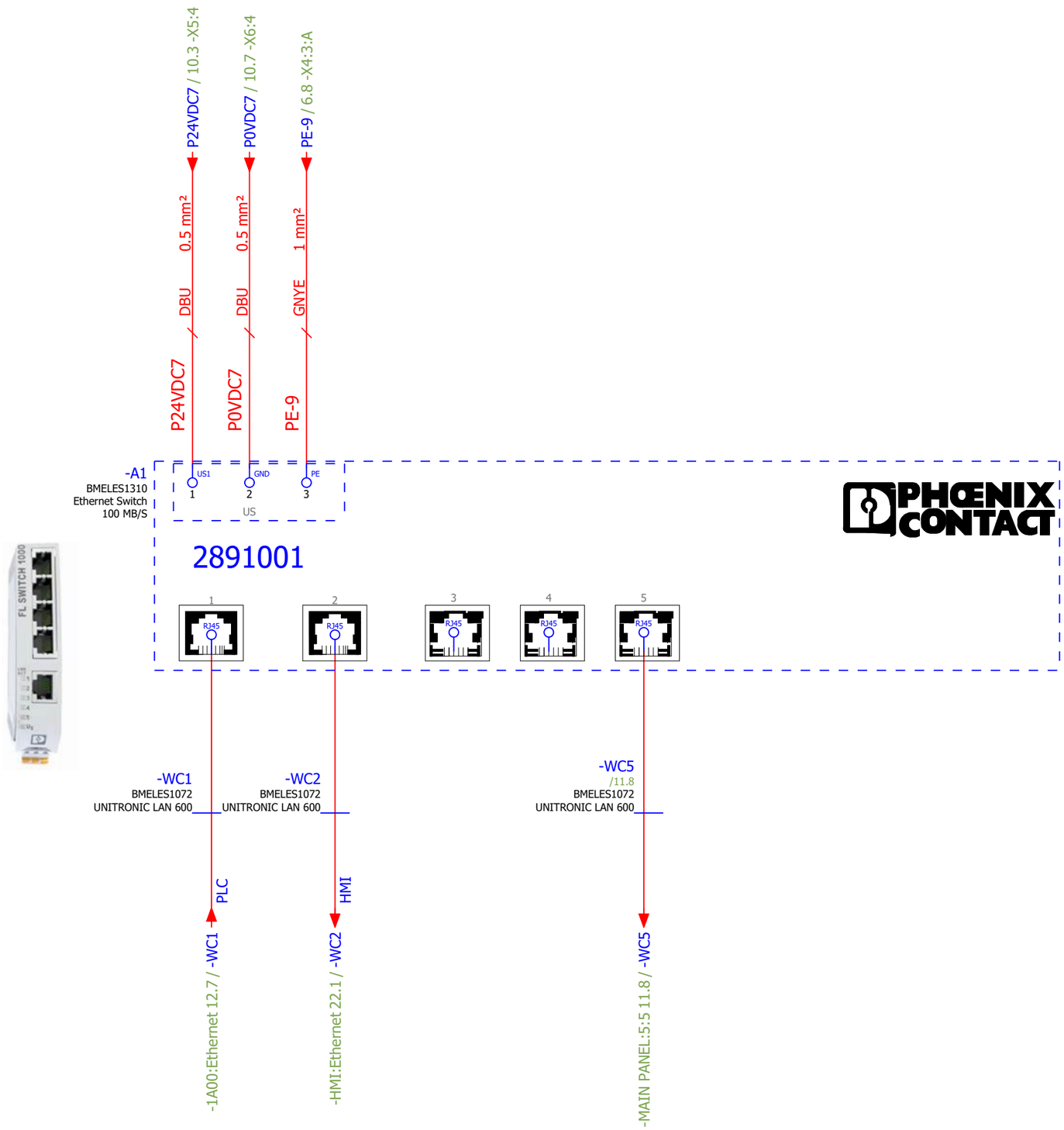
-X6
BMELES0364
TERMINAL BLOCK
2.5 SQ MM 1 IN 2 OUT

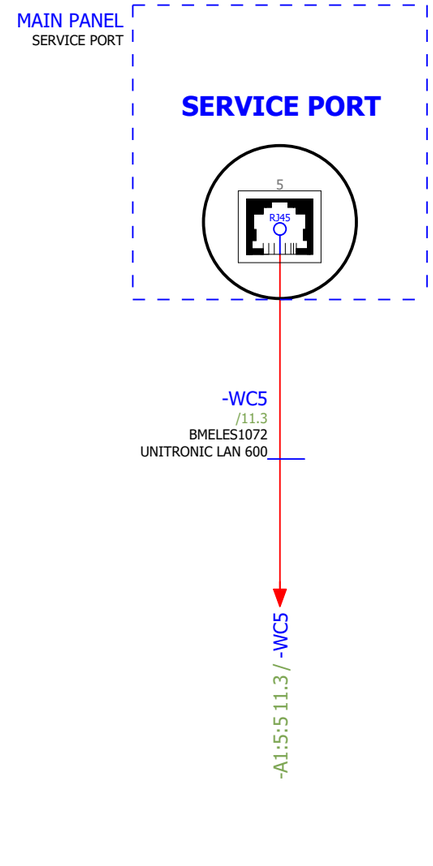
P0VDC1	DBU	0.5 mm²	-1A00:S/S 13.1 / P0VDC1
P0VDC2	DBU	0.5 mm²	-1A01:S/S 15.1 / P0VDC2
P0VDC3	DBU	0.5 mm²	-K6:A2 21.1 / P0VDC3
P0VDC4	DBU	0.5 mm²	-K1:A2 19.1 / P0VDC4
P0VDC5	BK/3	0.75 mm²	-PSC 22.5 / P0VDC5
P0VDC6	BK/2	0.5 mm²	-HMI:- 22.2 / P0VDC6
P0VDC7	DBU	0.5 mm²	-A1:US:2 11.1 / P0VDC7
P0VDC8	BK/17	0.5 mm²	+PP-1X1 +PP/2.1 / P0VDC8
P0VDC9	BK/5	0.5 mm²	-TL1.0:5 24.9 / P0VDC9
P0VDC10	BK/24	1 mm²	-1X100:21 25.7 / P0VDC10

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	DISTRIBUTION 24VDC TERMINAL BLOCK		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 10
Modification	Date	Name	Original		Replacement of	Replaced by			Page 11 / 32

COMMUNICATION







REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	COMMUNICATION		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 11
Modification	Date	Name	Original		Replacement of	Replaced by			Page 12 / 32

MAIN PLC POWER SUPPLY



-F6
BMELES1705
MCB SINGLE POLE
2A SIEMENS MAKE
5SL61027RC

1L1-7 / 6.3 -X2:2:A
1N1-7 / 6.6 -X3:7:B
PE-7 / 6.9 -X4:7:B

BK 2.5 mm²
1L1-7
2
1
2.5 mm² BK 2L1-7
2.5 mm² BU 1N1-7
2.5 mm² GNYE PE-7

WC1 / 11.1 -A1:1:1

-WC1
/11.1
/22.1
BMELES1072
2170144-UNITRONIC
LAN 600S/FTP CAT7Y



-1A00
CAAAZ00000000077
PLC
FX5U32MRES
MAKE :- MITSUBHISHI

L | N | PE | SG (GND) | SDB (TXD-) | SDA (TXD+) | RDB (RXD-) | RDA (RXD+) | V1+ (PEW0) | V2+ (PEW2) | V- (PEW4) | V+ (PAW0) | V- (PAW2) | Ethernet

Ethernet port

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 1 of 5

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN PLC POWER SUPPLY		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 12
Modification	Date	Name	Original		Replacement of	Replaced by			Page 13 / 32

MAIN INPUT MODULE



-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

P0VDC1 / 10.6 -X6:1

P0VDC1 — DBU — 0.5 mm²

S/S | 0V | 24V

X00	X01	X02	X03	X04	X05	X06	X07
X00 / 17.0 -X7:1:A	X01 / 17.0 -X7:1:C	X02 / 17.0 -X7:2:A	X03 / 17.1 -X7:2:C	X04 / 17.1 -X7:3:A	X05 / 17.1 -X7:3:C	X06 / 17.1 -X7:4:A	X07 / 17.1 -X7:4:C
DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU
0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²
E0.0	E0.1	E0.2	E0.3	E0.4	E0.5	E0.6	E0.7
EMERGENCY STOP PB	RH CYCLE START PB			AIR PRESSURE SWITCH		SCREW DRIVER OK ALL	
	LH CYCLE START PB		RESET		SPARE		SCREW DRIVER NG

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 2 of 5

-1A00

X10	X11	X12	X13	X14	X15	X16	X17
X10 / 17.1 -X7:5:A	X11 / 17.1 -X7:5:C	X12 / 17.2 -X7:6:A	X13 / 17.2 -X7:6:C	X14 / 17.2 -X7:7:A	X15 / 17.2 -X7:7:C	X16 / 17.2 -X7:8:A	X17 / 17.2 -X7:8:C
DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU
0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²
E1.0	E1.1	E1.2	E1.3	E1.4	E1.5	E1.6	E1.7
SPARE		CLAMPING CYL1 FWD RS		CLAMPING CYL2 FWD RS		LH MAIN PART PRESENCE	
	SPARE		CLAMPING CYL1 SPARE		CLAMPING CYL2 SPARE		LH CHILD PART PRESENCE

MITSUBISHI ELECTRIC
Mitsubishi FX5U-32MR-ES 16DI/16DO Part 3 of 5

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN INPUT MODULE		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 13
Modification	Date	Name	Original		Replacement of	Replaced by			Page 14 / 32

MAIN OUTPUT MODULE



-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

Mitsubishi FX5U-32MR-ES 16DI/16DO Part 4 of 5

MITSUBISHI ELECTRIC

A0.0	A0.1	A0.2	A0.3	A0.4	A0.5	A0.6	A0.7
TOWER LAMP GREEN	TOWER LAMP YELLOW	TOWER LAMP RED	TOWER LAMP BUZZER	SHUTOFF VALVE	MACHINE LIGHT	SCREW DRIVER ENABLE	SCREW DRIVER RESET
COM0	Y00	Y01	Y02	Y03	COM1	Y04	Y05	Y06	Y07

0.5 mm² DBU P24VDC2 → -X5:1 10.2 / P24VDC2

Y00	Y01	Y02	Y03	P24VDC2	Y04	Y05	Y06	Y07
-X8:1:A 18.1 / Y00	-X8:2:A 18.1 / Y01	-X8:3:A 18.1 / Y02	-X8:4:A 18.1 / Y03		-X8:5:A 18.2 / Y04	-K6:A1 21.0 / Y05	-K1:A1 19.0 / Y06	-K2:A1 19.2 / Y07

13

-1A00
CAAAZ00000000077
PLC FX5U32MRES
MAKE :- MITSUBHISHI

Mitsubishi FX5U-32MR-ES 16DI/16DO Part 5 of 5

MITSUBISHI ELECTRIC

A1.0	A1.1	A1.2	A1.3	A1.4	A1.5	A1.6	A1.7
SPARE	LH CLAMPING CYL FWD	LH CLAMPING CYL REV	RH CLAMPING CYL FWD	RH CLAMPING CYL REV	SPARE VALVE		SPARE VALVE
COM2	Y10	Y11	Y12	Y13	COM3	Y14	Y15	Y16	Y17

0.5 mm² DBU P24VDC3 → -X5:2 10.2 / P24VDC3

Y10	Y11	Y12	Y13	P24VDC3	Y14	Y15	Y16	Y17
-X8:6:A 18.2 / Y10	-X8:7:A 18.2 / Y11	-X8:8:A 18.2 / Y12	-X8:9:A 18.2 / Y13		-X8:10:A 18.3 / Y14	-X8:11:A 18.3 / Y15	-X8:12:A 18.3 / Y16	-X8:13:A 18.3 / Y17

15

REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	MAIN OUTPUT MODULE	=
			Ed.	MANORAJ.SELVARAJ				+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1		FMSPM00860	Page 14
Modification	Date	Name	Original		Replacement of	Replaced by		Page 15 / 32

EXPANSION INPUT MODULE



P0VDC2 / 10.6 -X6:1 — DBU 0.5 mm² — P0VDC2 — -1A01 S/S
X20 / 17.3 -X7:9:A — DBU 0.5 mm² — X20 — -1A01 X20 E2.0 — LH PROXIMITY SENSOR 1
X21 / 17.3 -X7:9:C — DBU 0.5 mm² — X21 — -1A01 X21 E2.1 — LH PROXIMITY SENSOR 2
X22 / 17.3 -X7:10:A — DBU 0.5 mm² — X22 — -1A01 X22 E2.2 — FIXTURE SPARE
X23 / 17.3 -X7:10:C — DBU 0.5 mm² — X23 — -1A01 X23 E2.3 — FIXTURE SPARE
X24 / 17.3 -X7:11:A — DBU 0.5 mm² — X24 — -1A01 X24 E2.4 — FIXTURE SPARE
X25 / 17.3 -X7:11:C — DBU 0.5 mm² — X25 — -1A01 X25 E2.5 — FIXTURE SPARE
X26 / 17.3 -X7:12:A — DBU 0.5 mm² — X26 — -1A01 X26 E2.6 — HIGH VARIANT
X27 / 17.4 -X7:12:C — DBU 0.5 mm² — X27 — -1A01 X27 E2.7 — MID VARIANT

-1A01
/15.5

CAAAZ00000000072
16POINTINPUTCARD
MODEL :-FX5-16EX/ES
MAKE :-MITSUBHISHI

MITSUBISHI ELECTRIC
Mitsubishi FX5U-EX/ES 16DI Part 1 of 2

X30 / 17.4 -X7:13:A — DBU 0.5 mm² — X30 — -1A01 X30 E3.0 — LOW VARIANT
X31 / 17.4 -X7:13:C — DBU 0.5 mm² — X31 — -1A01 X31 E3.1 — SPARE
X32 / 17.4 -X7:14:A — DBU 0.5 mm² — X32 — -1A01 X32 E3.2 — SPARE
X33 / 17.4 -X7:14:C — DBU 0.5 mm² — X33 — -1A01 X33 E3.3 — SPARE
X34 / 17.4 -X7:15:A — DBU 0.5 mm² — X34 — -1A01 X34 E3.4 — SPARE
X35 / 17.5 -X7:15:C — DBU 0.5 mm² — X35 — -1A01 X35 E3.5 — SPARE
X36 / 17.5 -X7:16:A — DBU 0.5 mm² — X36 — -1A01 X36 E3.6 — SPARE
X37 / 17.5 -X7:16:C — DBU 0.5 mm² — X37 — -1A01 X37 E3.7 — SPARE

-1A01
/15.1

MITSUBISHI ELECTRIC
Mitsubishi FX5U-EX/ES 16DI Part 2 of 2

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	EXPANSION INPUT MODULE		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 15
Modification	Date	Name	Original		Replacement of	Replaced by			Page 16 / 32

EXPANSION OUTPUT MODULE



-1A02
/16.5
CAAAZ00000000073
16POINTS OUTCARD
-FX5-16EYRES, MAKE
MISTUBHISHI

Mitsubishi FX5-16EYR-ES 16DO Part 1 of 2

MITSUBISHI ELECTRIC

SPARE VALVE SPARE SPARE SPARE

SPARE VALVE SPARE SPARE SPARE

+V4 Y20 Y21 Y22 Y23 Y24 Y25 Y26 Y27

0.5 mm² DBU

P24VDC4

Y20 Y21 Y22 Y23 Y24 Y25 Y26 Y27

-X8:14:A 18.4 / Y20
-X8:15:A 18.4 / Y21
-X8:16:A 18.4 / Y22
-X8:17:A 18.4 / Y23
-X8:18:A 18.4 / Y24
-X8:19:A 18.5 / Y25
-X8:20:A 18.5 / Y26
-X8:21:A 18.5 / Y27

-X5:2 10.3 / P24VDC4

-1A02
/16.1

Mitsubishi FX5-16EYR-ES 16DO Part 2 of 2

MITSUBISHI ELECTRIC

SPARE SPARE SPARE SPARE

SPARE SPARE SPARE SPARE

+V6 Y30 Y31 Y32 Y33 Y34 Y35 Y36 Y37

0.5 mm² DBU

P24VDC5

Y30 Y31 Y32 Y33 Y34 Y35 Y36 Y37

-X8:22:A 18.5 / Y30
-X8:23:A 18.5 / Y31
-X8:24:A 18.6 / Y32
-X8:25:A 18.6 / Y33
-X8:26:A 18.6 / Y34
-X8:27:A 18.6 / Y35
-X8:28:A 18.7 / Y36
-X8:29:A 18.7 / Y37

-X5:3 10.3 / P24VDC5

15 17

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	EXPANSION OUTPUT MODULE	=
			Ed.	manoraj.selvaraj				+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860 Page 16
Modification	Date	Name	Original		Replacement of	Replaced by		Page 17 / 32

BMELES2970
TERMINAL BLOCK CXDL2.5
2.5 SQ MM 2 IN 2 OUT
-X7-

-PB1-S1:12 24.5 / X00	X00	BK/4	0.75 mm²	1	X00	DBU	0.5 mm²	X00 / 13.2 -1A00:X00
-PB2-S4:14 24.1 / X01	X01	BK/3	0.75 mm²		X01	DBU	0.5 mm²	X01 / 13.3 -1A00:X01
-PB1-S2:14 24.4 / X02	X02	BK/3	0.75 mm²	2	X02	DBU	0.5 mm²	X02 / 13.3 -1A00:X02
-PB2-S3:14 24.2 / X03	X03	BK/4	0.75 mm²		X03	DBU	0.5 mm²	X03 / 13.3 -1A00:X03
-PSC 22.6 / X04	X04	BK/2	0.75 mm²	3	X04	DBU	0.5 mm²	X04 / 13.4 -1A00:X04
					X05	DBU	0.5 mm²	X05 / 13.4 -1A00:X05
-K4:14 19.7 / X06	X06	DBU	0.5 mm²	4	X06	DBU	0.5 mm²	X06 / 13.4 -1A00:X06
-K5:14 19.9 / X07	X07	DBU	0.5 mm²		X07	DBU	0.5 mm²	X07 / 13.5 -1A00:X07
				5	X10	DBU	0.5 mm²	X10 / 13.6 -1A00:X10
					X11	DBU	0.5 mm²	X11 / 13.6 -1A00:X11
-1X100:1 25.1 / X12	X12	BK/1	1 mm²	6	X12	DBU	0.5 mm²	X12 / 13.7 -1A00:X12
-1X100:2 25.1 / X13	X13	BK/2	1 mm²		X13	DBU	0.5 mm²	X13 / 13.7 -1A00:X13
-1X100:3 25.1 / X14	X14	BK/3	1 mm²	7	X14	DBU	0.5 mm²	X14 / 13.8 -1A00:X14
-1X100:4 25.2 / X15	X15	BK/4	1 mm²		X15	DBU	0.5 mm²	X15 / 13.8 -1A00:X15
-1X100:5 25.2 / X16	X16	BK/5	1 mm²	8	X16	DBU	0.5 mm²	X16 / 13.8 -1A00:X16
-1X100:6 25.2 / X17	X17	BK/6	1 mm²		X17	DBU	0.5 mm²	X17 / 13.9 -1A00:X17
-1X100:7 25.3 / X20	X20	BK/7	1 mm²	9	X20	DBU	0.5 mm²	X20 / 15.1 -1A01:X20
-1X100:8 25.3 / X21	X21	BK/8	1 mm²		X21	DBU	0.5 mm²	X21 / 15.2 -1A01:X21
-1X100:9 25.3 / X22	X22	BK/9	1 mm²	10	X22	DBU	0.5 mm²	X22 / 15.2 -1A01:X22
-1X100:10 25.3 / X23	X23	BK/10	1 mm²		X23	DBU	0.5 mm²	X23 / 15.3 -1A01:X23
-1X100:11 25.4 / X24	X24	BK/11	1 mm²	11	X24	DBU	0.5 mm²	X24 / 15.3 -1A01:X24
-1X100:12 25.4 / X25	X25	BK/12	1 mm²		X25	DBU	0.5 mm²	X25 / 15.3 -1A01:X25
-1X100:13 25.4 / X26	X26	BK/13	1 mm²	12	X26	DBU	0.5 mm²	X26 / 15.4 -1A01:X26
-1X100:14 25.5 / X27	X27	BK/14	1 mm²		X27	DBU	0.5 mm²	X27 / 15.4 -1A01:X27
-1X100:15 25.5 / X30	X30	BK/15	1 mm²	13	X30	DBU	0.5 mm²	X30 / 15.6 -1A01:X30
					X31	DBU	0.5 mm²	X31 / 15.6 -1A01:X31
				14	X32	DBU	0.5 mm²	X32 / 15.6 -1A01:X32
					X33	DBU	0.5 mm²	X33 / 15.7 -1A01:X33
				15	X34	DBU	0.5 mm²	X34 / 15.7 -1A01:X34
					X35	DBU	0.5 mm²	X35 / 15.7 -1A01:X35
				16	X36	DBU	0.5 mm²	X36 / 15.8 -1A01:X36
					X37	DBU	0.5 mm²	X37 / 15.8 -1A01:X37

INPUT TERMINAL DISTRIBUTION

Modification	REVISION			
Date				
Name				
Original	Appr.	Ed.	Date	01-08-2023
		manoraj.selvaraj		
Replacement of	ROBIS			
	S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			
Replaced by	Motherson Automotive & Technology engineering			
	INPUT TERMINAL DISTRIBUTION			
	FMSPM00860			

OUTPUT TERMINAL DISTRIBUTION

ALL FUSE SHOULD BE 0.5 A



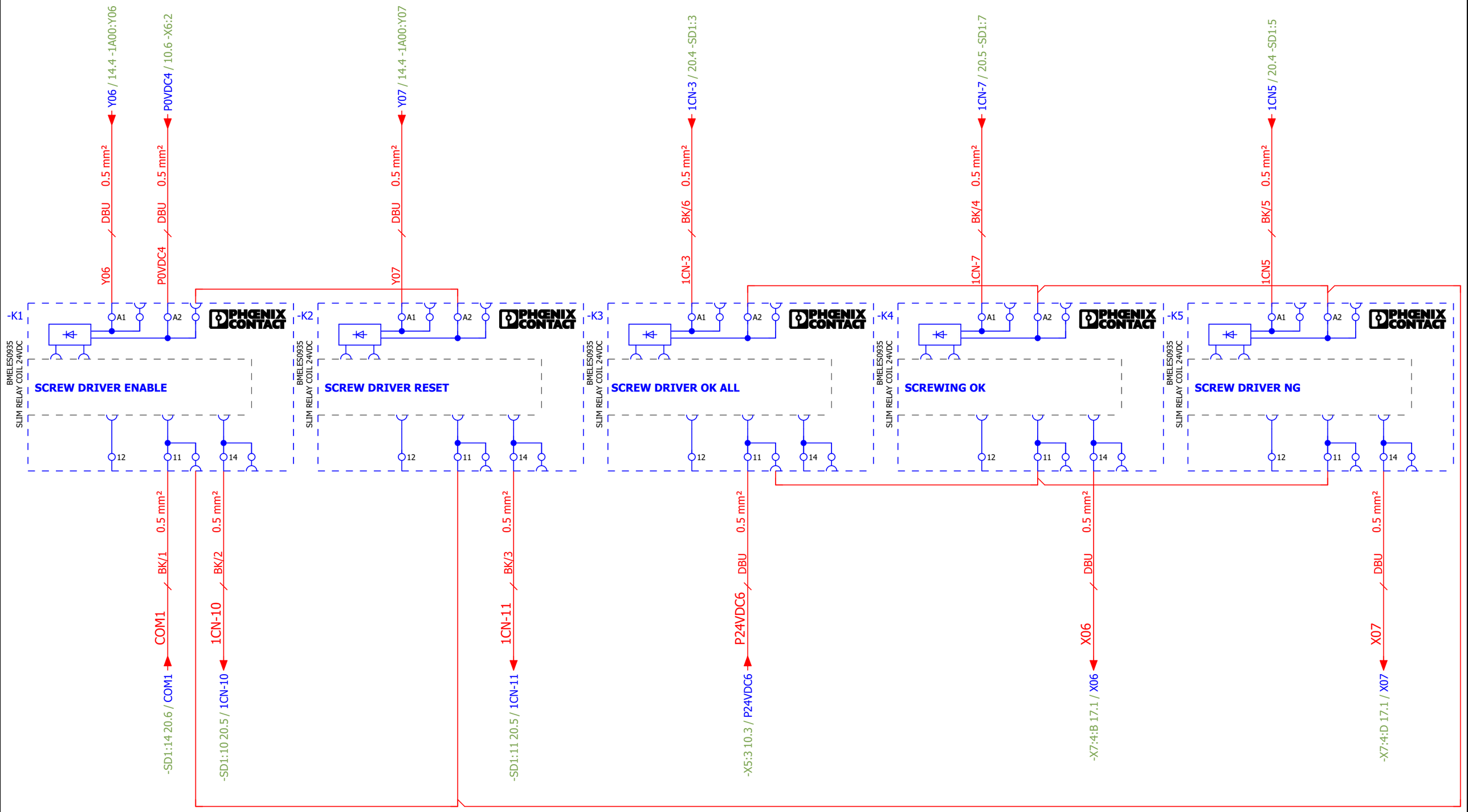
-X8
BMELES2968
TERMINAL BLOCK 2.5 SQ MM
1 IN 2 OUT

Y00 / 14.1 / -1A00:Y00	Y01 / 14.2 / -1A00:Y01	Y02 / 14.2 / -1A00:Y02	Y03 / 14.3 / -1A00:Y03	Y04 / 14.3 / -1A00:Y04	Y10 / 14.6 / -1A00:Y10	Y11 / 14.6 / -1A00:Y11	Y12 / 14.7 / -1A00:Y12	Y13 / 14.7 / -1A00:Y13	Y14 / 14.8 / -1A00:Y14	Y15 / 14.8 / -1A00:Y15	Y16 / 14.8 / -1A00:Y16	Y17 / 14.9 / -1A00:Y17	Y20 / 16.1 / -1A02:Y20	Y21 / 16.2 / -1A02:Y21	Y22 / 16.2 / -1A02:Y22	Y23 / 16.3 / -1A02:Y23	Y24 / 16.3 / -1A02:Y24	Y25 / 16.3 / -1A02:Y25	Y26 / 16.4 / -1A02:Y26	Y27 / 16.4 / -1A02:Y27	Y30 / 16.6 / -1A02:Y30	Y31 / 16.6 / -1A02:Y31	Y32 / 16.6 / -1A02:Y32	Y33 / 16.7 / -1A02:Y33	Y34 / 16.7 / -1A02:Y34	Y35 / 16.8 / -1A02:Y35	Y36 / 16.8 / -1A02:Y36	Y37 / 16.8 / -1A02:Y37
0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²
DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU	DBU
Y00	Y01	Y02	Y03	Y04	Y10	Y11	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21	Y22	Y23	Y24	Y25	Y26	Y27	Y30	Y31	Y32	Y33	Y34	Y35	Y36	Y37		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29		
0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A	0.5A		
OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE	OE		

Y00	Y01	Y02	Y03	Y04	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21																		
0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²	0.5 mm²																		
BK/1	BK/2	BK/3	BK/4	BK/1	BK/2	BK/3	BK/4	BK/5	BK/6	BK/7	BK/8	BK/9																		
Y00	Y01	Y02	Y03	Y04	Y12	Y13	Y14	Y15	Y16	Y17	Y20	Y21																		
-TL1.0:1 24.7 / Y00	-TL1.0:2 24.8 / Y01	-TL1.0:3 24.8 / Y02	-TL1.0:4 24.8 / Y03	+PP-1X1:1 +PP/2.1 / Y04	+PP-1X1:2 +PP/2.1 / Y12	+PP-1X1:3 +PP/2.2 / Y13	+PP-1X1:4 +PP/2.2 / Y14	+PP-1X1:5 +PP/2.2 / Y15	+PP-1X1:6 +PP/2.3 / Y16	+PP-1X1:7 +PP/2.3 / Y17	+PP-1X1:8 +PP/2.3 / Y20	+PP-1X1:9 +PP/2.4 / Y21																		

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	OUTPUT TERMINAL DISTRIBUTION		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 18
Modification	Date	Name	Original		Replacement of	Replaced by			Page 19 / 32

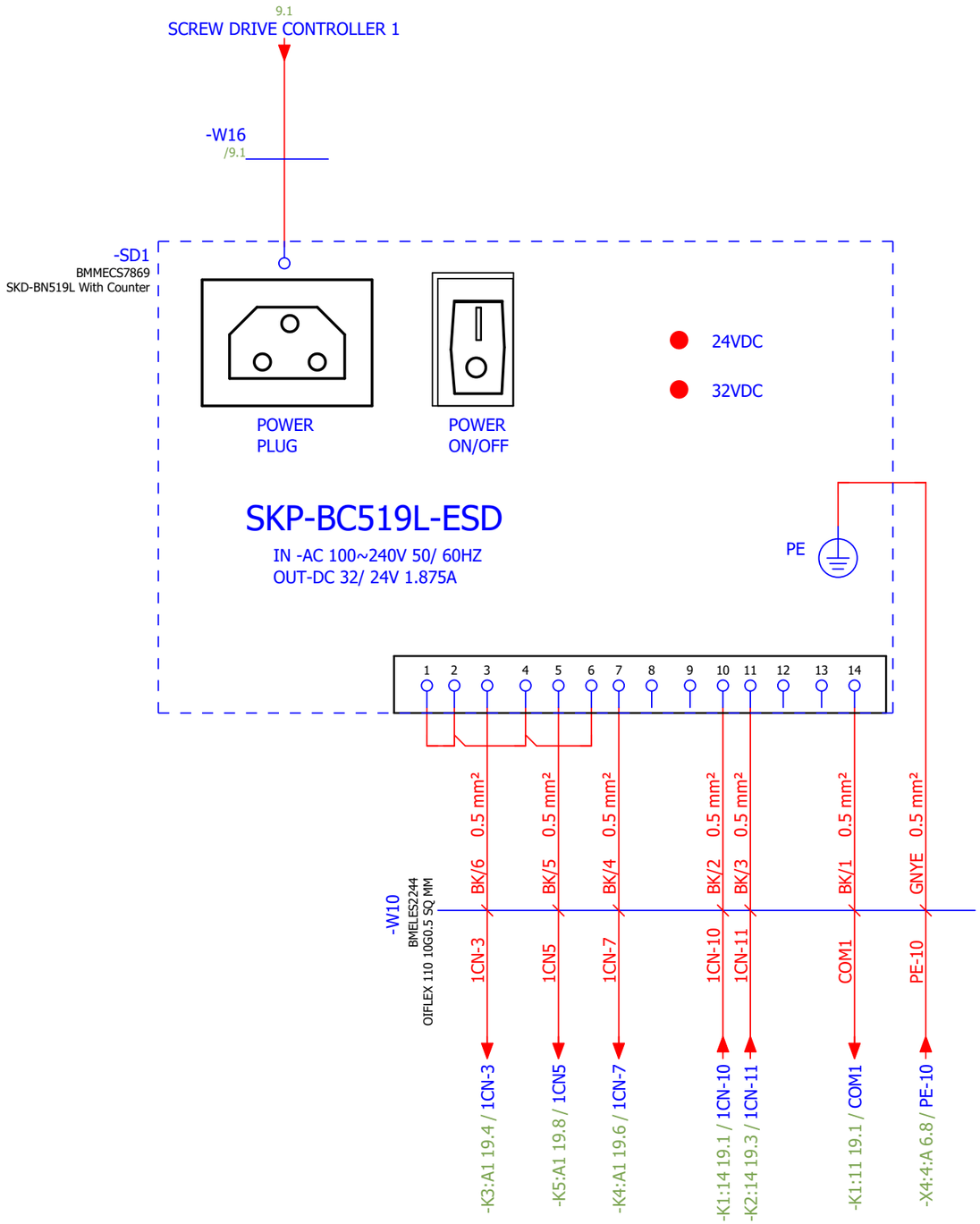
RELAY INTERFACE FOR SCREWING DRIVER 1



18 20

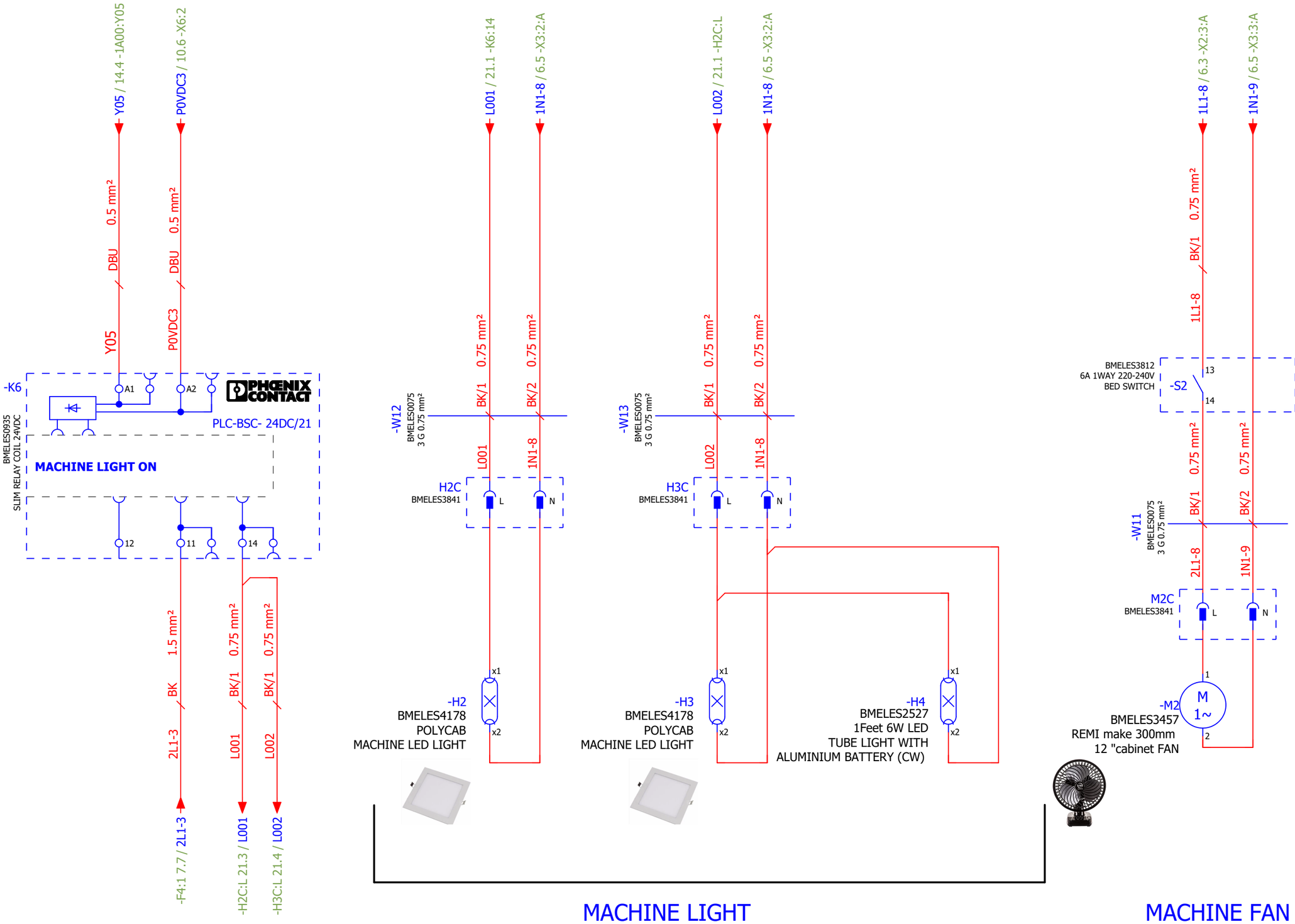
REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	RELAY INTERFACE FOR SCREWING DRIVER 1		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 19
Modification	Date	Name	Original		Replacement of	Replaced by			Page 20 / 32

SCREWING DRIVER INTERFACE 1



REVISION			Date	11-04-2024	ROBIS	Motherson Automotive & Technology engineering	SCREWING DRIVER INTERFACE 1		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 20
Modification	Date	Name	Original		Replacement of	Replaced by			Page 21 / 32

MACHINE LIGHT & FAN INTERFACE



MACHINE LIGHT

MACHINE FAN

20

22

REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	MACHINE LIGHT & FAN INTERFACE		=
			Ed.	MANORAJ.SELVARAJ	S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1				+ CP
			Appr.					FMSPM00860	Page 21
Modification	Date	Name	Original		Replacement of	Replaced by			Page 22 / 32

HMI INTERFACE



-WC2 / 11.2 -A1:2:2

-WC6 / 23.2 -HMI RS232-COM1

-WC1
BMELES1072
UNITRONIC LAN 600

-WC6
BMELES0221
0034504 - Unitronic
LIYCY 4 X 0.34

-HMI
CAAAI00000000586
HMI GS2107-WTBD

Ethernet | RS-232 | RS-422 | USB

MITSUBISHI ELECTRIC

\+ | - | FG

0.75 mm² | 0.75 mm² | 0.75 mm²

-W5
BMELES1937
3 G 0.75 mm²

BK/1 | BK/2 | GNYE

1P24VDC | P0VDC6 | PE-8

-1F1 / 10.0 / 1P24VDC

-X6:3 / 10.7 / P0VDC6

-X4:2:A / 6.7 / PE-8

HMI POWER SUPPLY

P24VDC12 / 10.4 -X5:6

P0VDC5 / 10.7 -X6:3

X04 / 17.1 -X7:3:B

0.75 mm² | 0.75 mm² | 0.75 mm²

-W6
BMELES1937
3 G 0.75 mm²

BK/1 | BK/3 | BK/2

P24VDC12 | P0VDC5 | X04

-PSC
3 PIN CONNECTOR

BN | BU | BK | WH

PS1

1 | 2 | 3 | 4

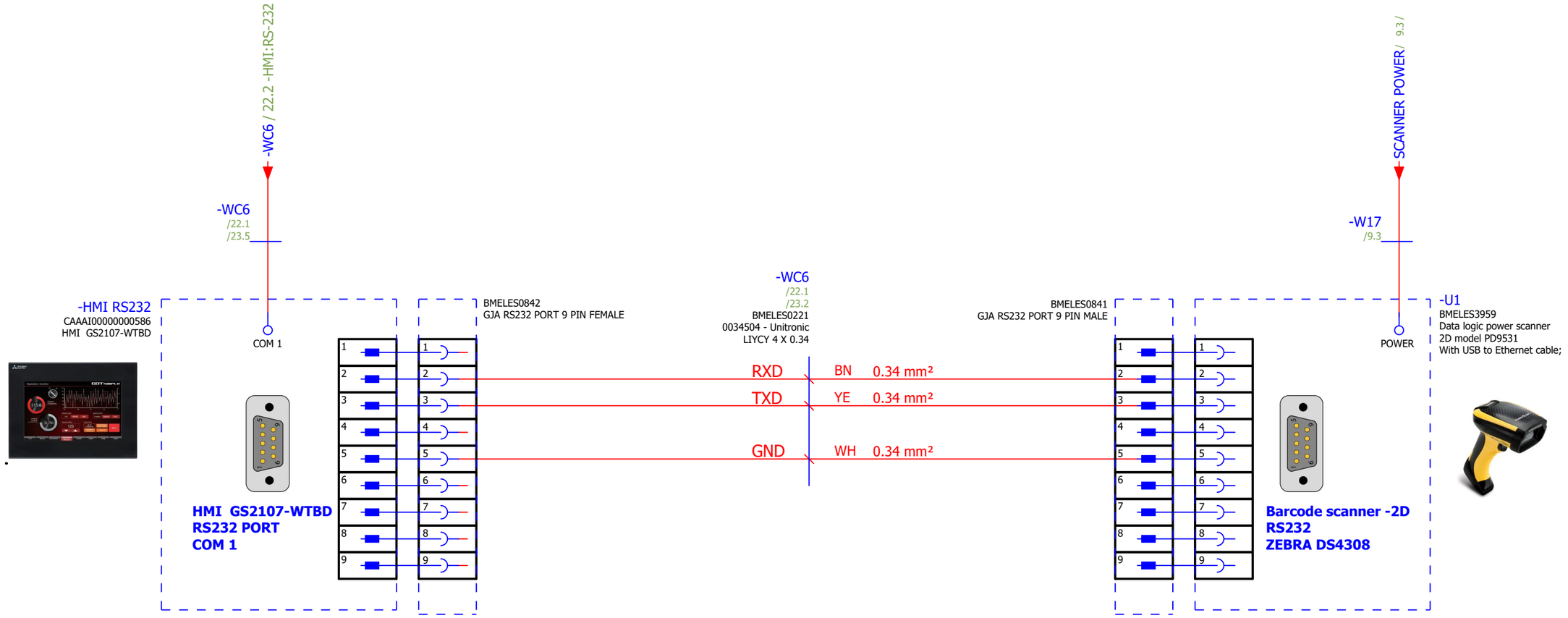
P

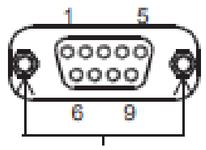


PRESSURE SWITCH

REVISION			Date	24-07-2023	ROBIS	Motherson Automotive & Technology engineering	HMI INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 22
Modification	Date	Name	Original		Replacement of	Replaced by			Page 23 / 32

HMI TO SCANNER COMMUNICATION



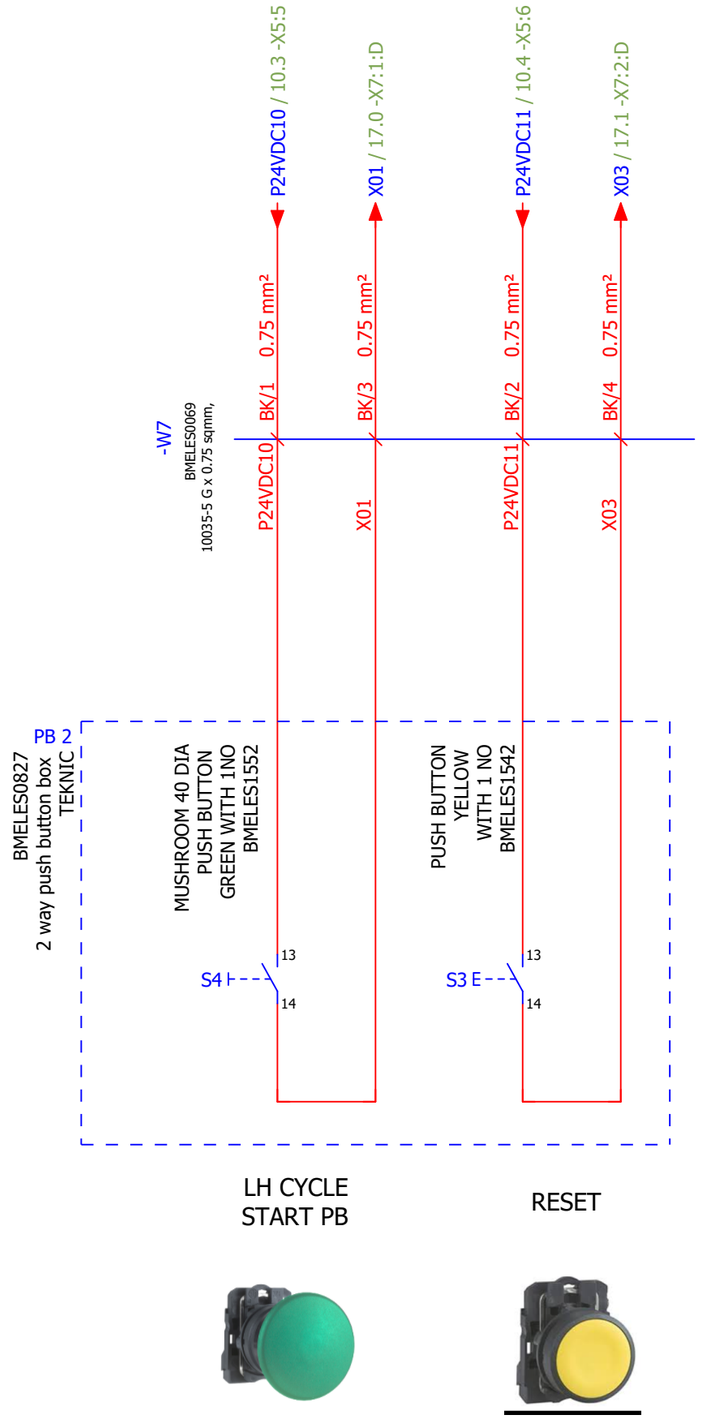


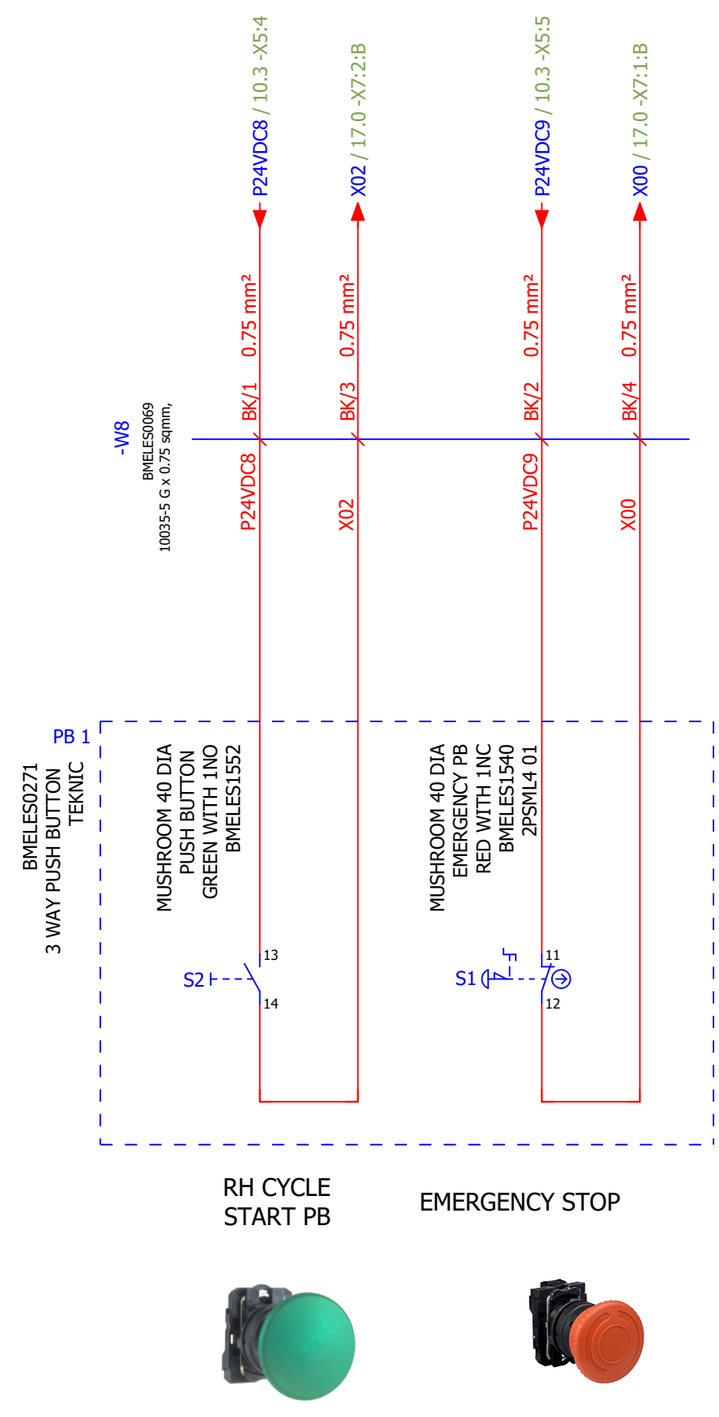
Screws to fix
RS-232C connector
#4-40UNC
(inch screw thread)

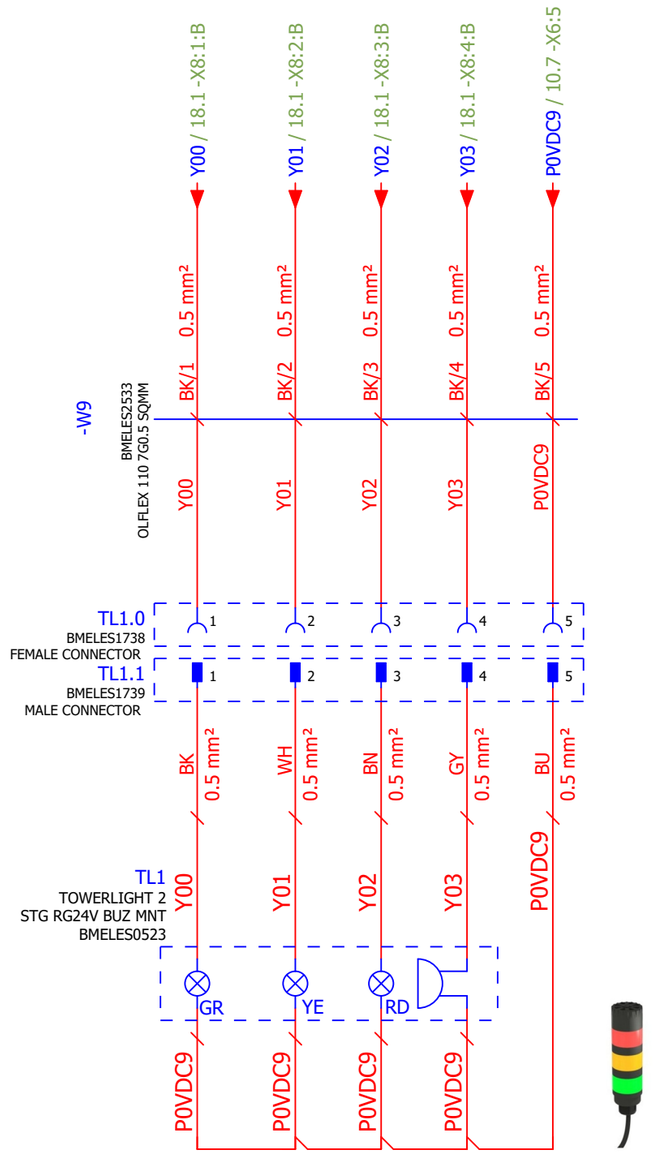
Pin No.	Signal	Name
1	CD (DCD)	Receive carrier detection
2	RD (RXD)	Receive data
3	SD (TXD)	Send data
4	ER (DTR)	Send request
5	SG (GND)	Signal ground
6	DR (DSR)	Send enabled
7, 8, 9		Not used

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	HMI TO SCANNER COMMUNICATION		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 23
Modification	Date	Name	Original		Replacement of	Replaced by			Page 24 / 32

PUSHBUTTON & TOWER LAMP INTERFACE

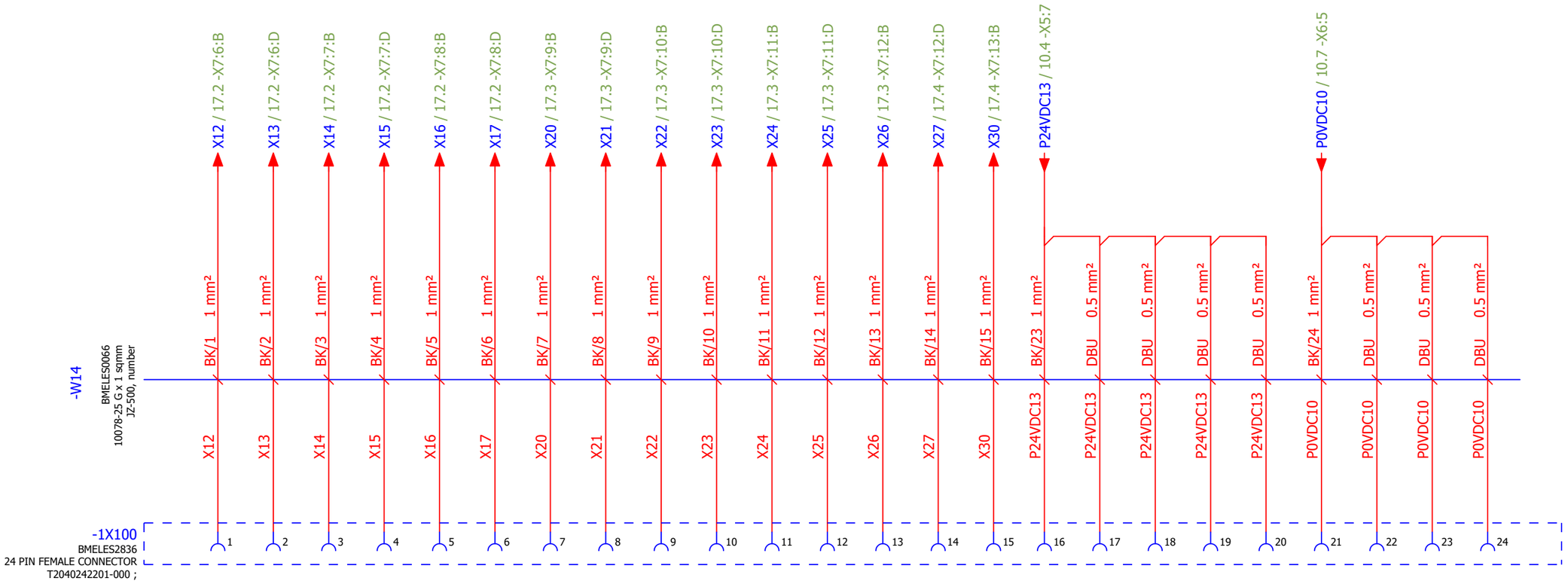






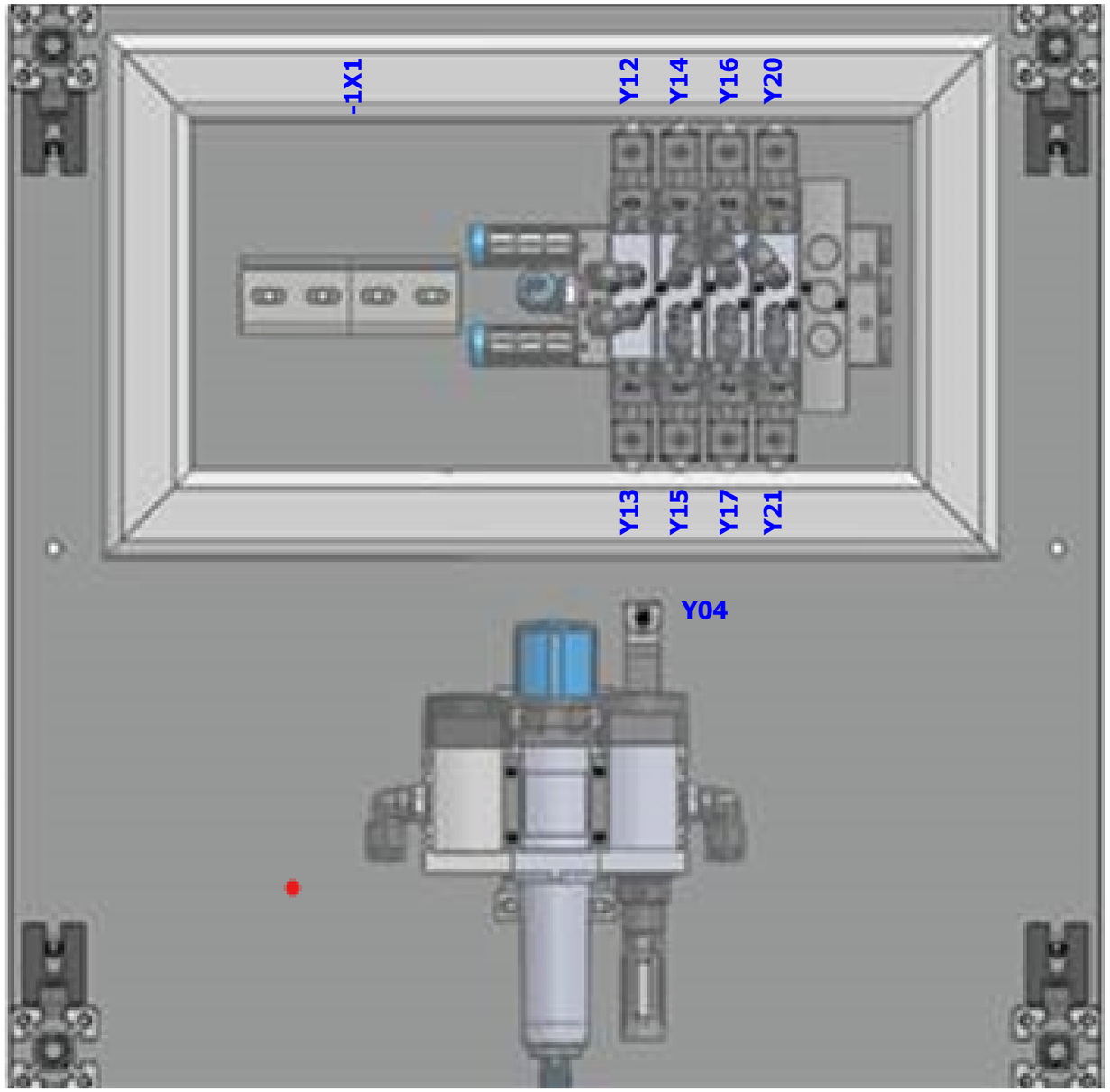
REVISION			Date	07-09-2023	ROBIS	Motherson Automotive & Technology engineering	PUSHBUTTON & TOWER LAMP INTERFACE		=
			Ed.	MANORAJ.SELVARAJ					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 24
Modification	Date	Name	Original		Replacement of	Replaced by			Page 25 / 32

24 PIN I/O INTERFACE



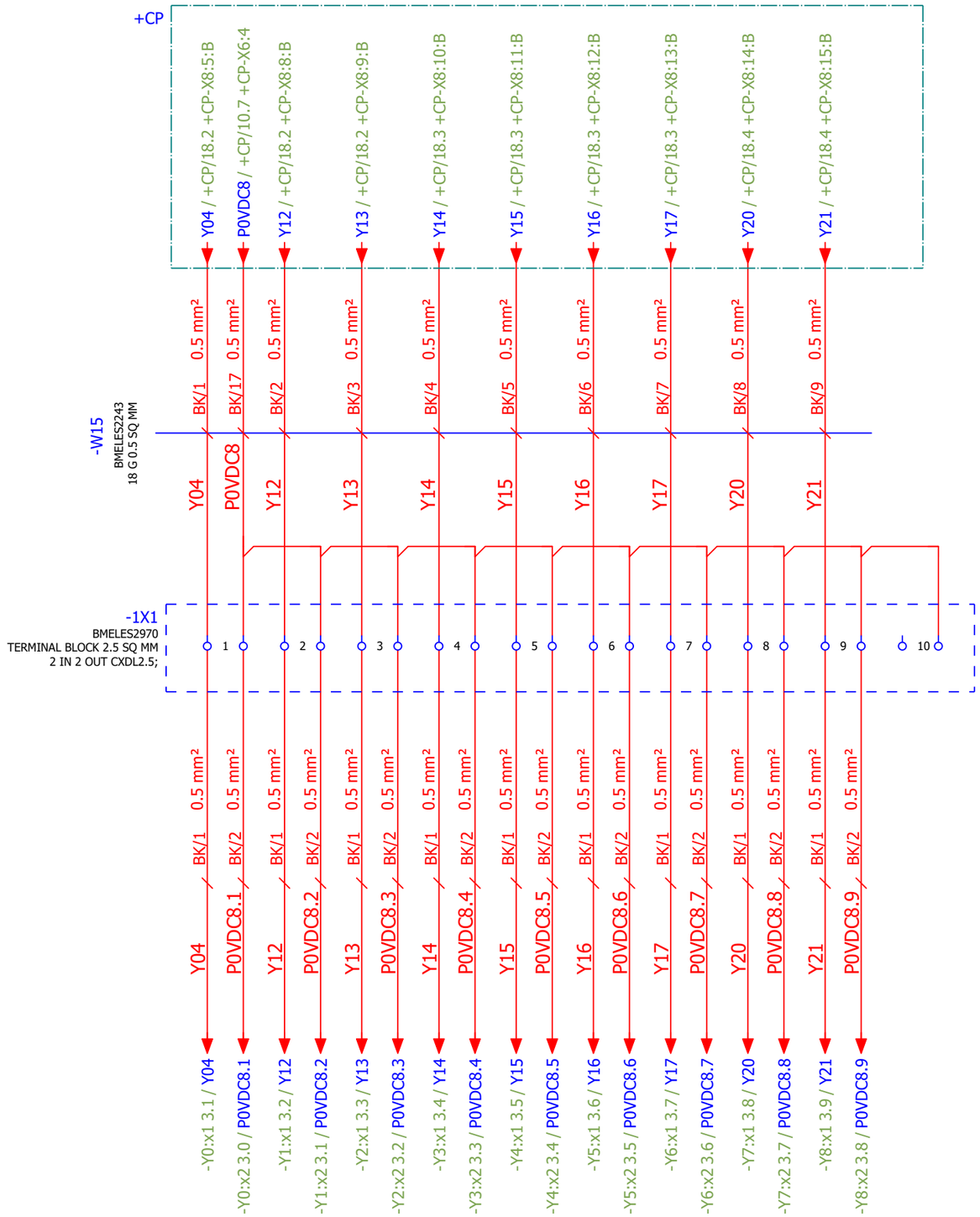
REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	24 PIN I/O INTERFACE		=
			Ed.	manoraj.selvaraj					+ CP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 25
Modification	Date	Name	Original		Replacement of	Replaced by			Page 26 / 32

CUSTOMIZE PANEL



TAG NO	DESCRIPTION
Y04	SHUT OFF VALVE
Y12	CLAMPING CYL1 FWD SOL
Y13	CLAMPING CYL1 REV SOL
Y14	SPARE VALVE
Y15	SPARE VALVE
Y16	SPARE VALVE
Y17	SPARE VALVE
Y20	SPARE VALVE
Y21	SPARE VALVE
-1X1	BMELES2970 TERMINAL BLOCK 2.5 SQ MM 2 IN 2 OUT CXDL2.5;

PNEUMATIC TERMINAL BLOCK



REVISION			Date	07-09-2023	ROBIS		Motherson Automotive & Technology engineering	PNEUMATIC TERMINAL BLOCK	=
			Ed.	MANORAJ.SELVARAJ					+ PP
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 2
Modification	Date	Name	Original		Replacement of	Replaced by			Page 28 / 32

REVISION			
Modification	Date	Name	Original
		Ed.	Date
		Appr.	07-09-2023
		MANORAJ.SELVARAJ	
Replacement of			Replaced by

ROBIS

S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1

Motherson Automotive & Technology engineering

CYLINDER INTERFACE

FMSPM00860

+ PP

=

Page 29 / 32

3

+FIX/1

CYLINDER INTERFACE

SHUT OFF VALVE

Y04
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.1 BK/2 0.5 mm² → P0VDC8.1 / 2.1 -1X1
-Y0 x2 x1
Y04 BK/1 0.5 mm² → Y04 / 2.1 -1X1:1

CLAMP CYL 1 FWD SOL

Y12
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.2 BK/2 0.5 mm² → P0VDC8.2 / 2.1 -1X1
-Y1 x2 x1
Y12 BK/1 0.5 mm² → Y12 / 2.1 -1X1:2

CLAMP CYL 1 REV SOL

Y13
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.3 BK/2 0.5 mm² → P0VDC8.3 / 2.2 -1X1
-Y2 x2 x1
Y13 BK/1 0.5 mm² → Y13 / 2.2 -1X1:3

CLAMP CYL 2 FWD SOL

Y14
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.4 BK/2 0.5 mm² → P0VDC8.4 / 2.2 -1X1
-Y3 x2 x1
Y14 BK/1 0.5 mm² → Y14 / 2.2 -1X1:4

CLAMP CYL 2 REV SOL

Y15
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.5 BK/2 0.5 mm² → P0VDC8.5 / 2.3 -1X1
-Y4 x2 x1
Y15 BK/1 0.5 mm² → Y15 / 2.2 -1X1:5

SPARE VALVE

Y16
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.6 BK/2 0.5 mm² → P0VDC8.6 / 2.3 -1X1
-Y5 x2 x1
Y16 BK/1 0.5 mm² → Y16 / 2.3 -1X1:6

SPARE VALVE

Y17
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.7 BK/2 0.5 mm² → P0VDC8.7 / 2.3 -1X1
-Y6 x2 x1
Y17 BK/1 0.5 mm² → Y17 / 2.3 -1X1:7

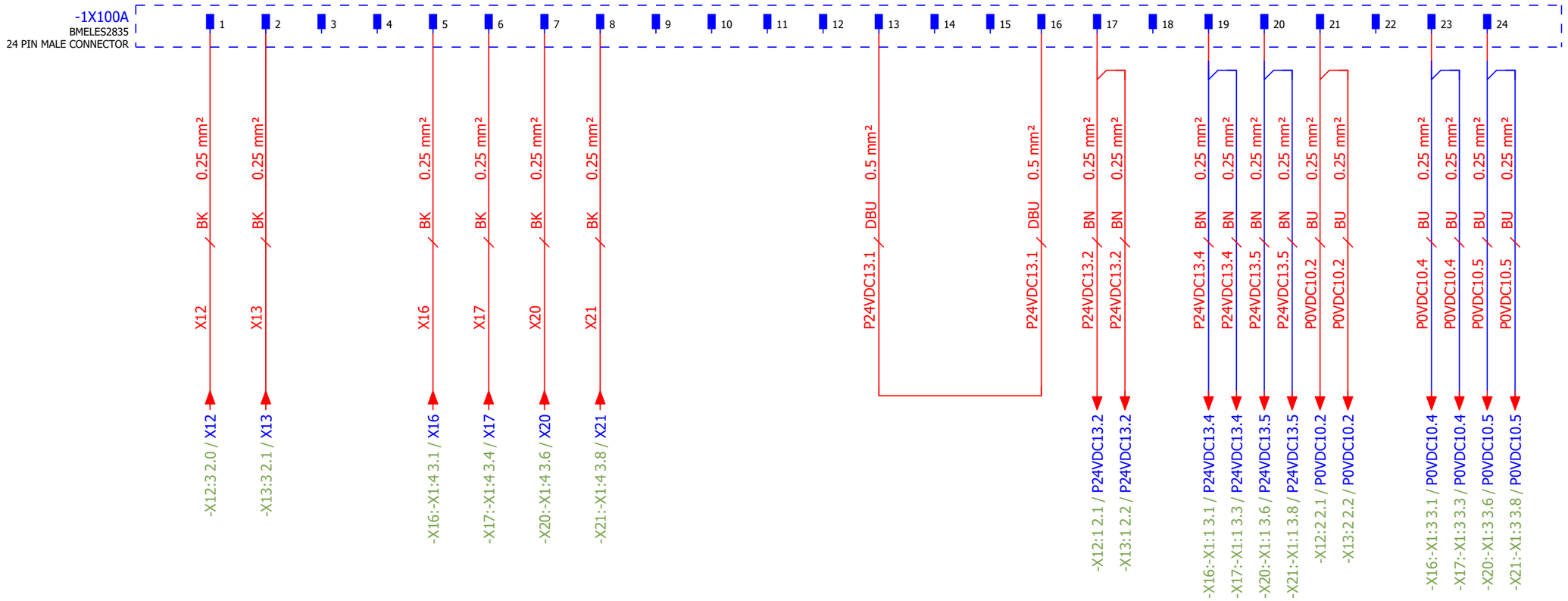
SPARE VALVE

Y20
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.8 BK/2 0.5 mm² → P0VDC8.8 / 2.4 -1X1
-Y7 x2 x1
Y20 BK/1 0.5 mm² → Y20 / 2.3 -1X1:8

SPARE VALVE

Y21
BMPNES2070
KMEB-1-24-2.5-LED
P0VDC8.9 BK/2 0.5 mm² → P0VDC8.9 / 2.4 -1X1
-Y8 x2 x1
Y21 BK/1 0.5 mm² → Y21 / 2.4 -1X1:9

24 PIN I/O INTERFACE

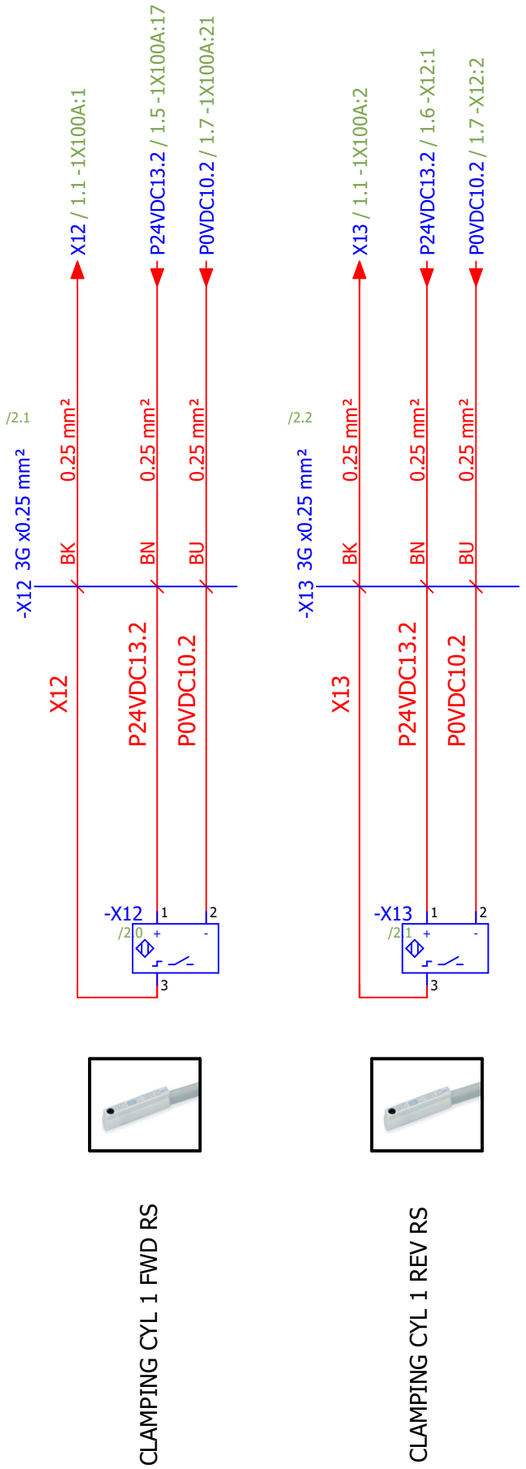


+PP/3

2

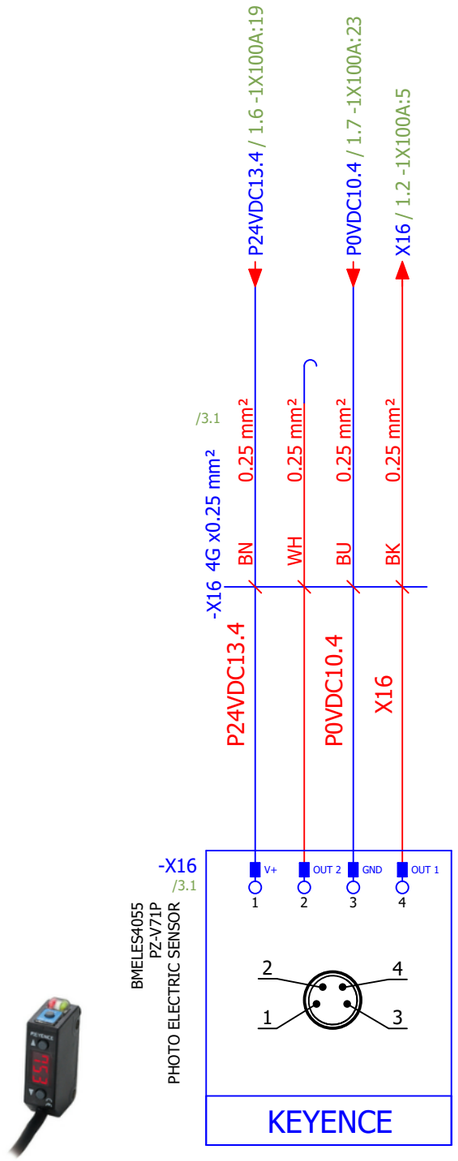
REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	24 PIN I/O INTERFACE	=
			Ed.	manoraj.selvaraj				+ FIX
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860 Page 1
Modification	Date	Name	Original		Replacement of	Replaced by		Page 30 / 32

REED SWITCH INTERFACE

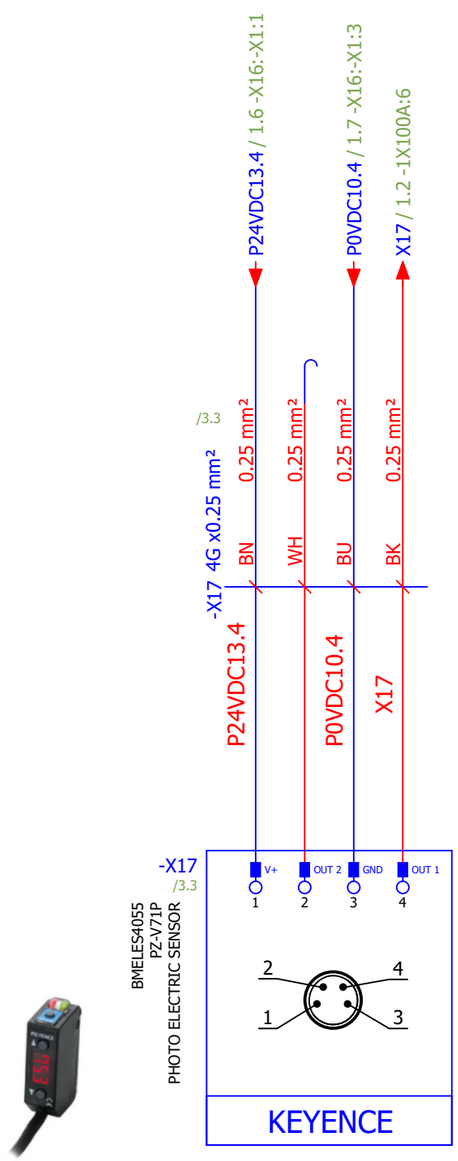


REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	REED SWITCH INTERFACE		=
			Ed.	manoraj.selvaraj					+ FIX
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 2
Modification	Date	Name	Original		Replacement of	Replaced by			Page 31 / 32

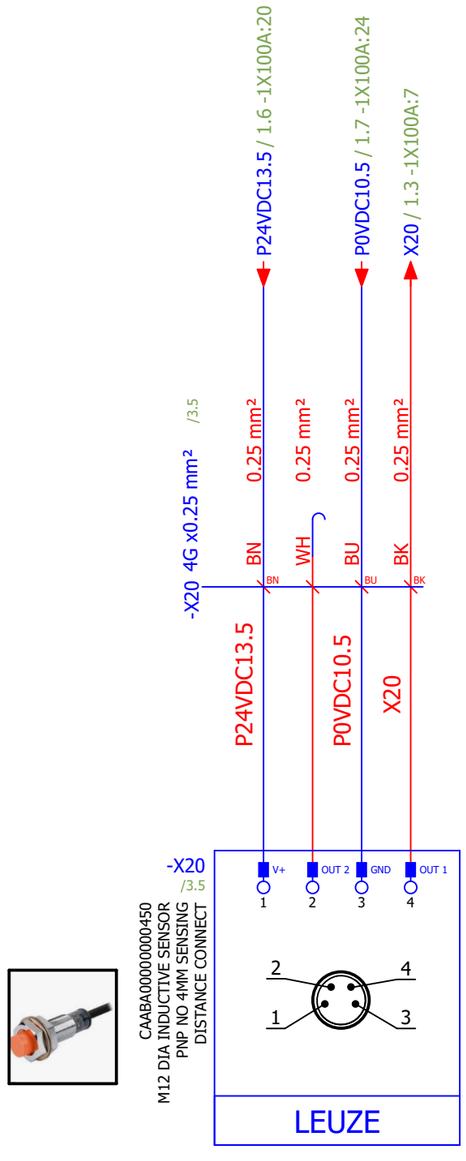
SENSOR INTERFACE



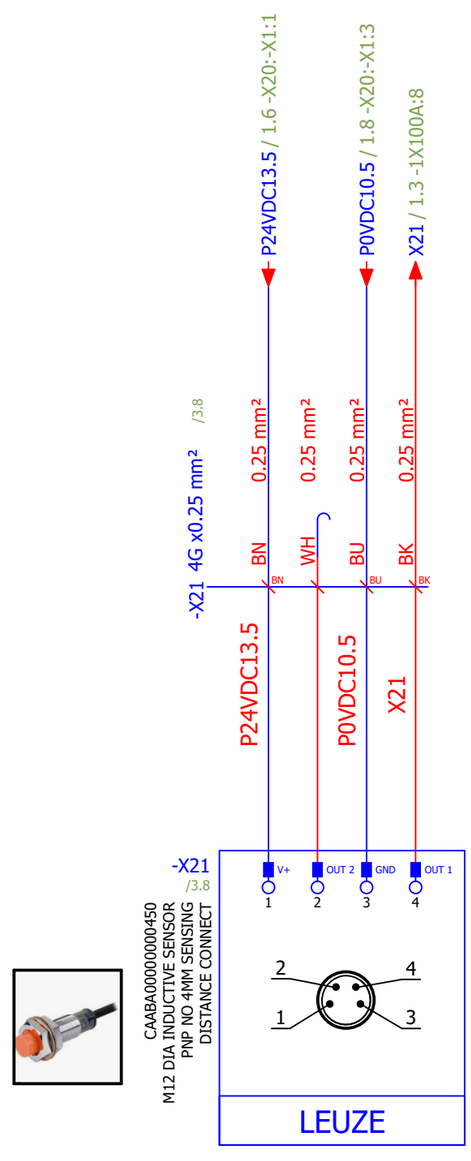
MAIN PART PRESENCE SENOSR



CHILD PART PRESENCE SENOSR



PROXIMITY SENSOR 1



PROXIMITY SENSOR 2

2

REVISION			Date	01-08-2023	ROBIS	Motherson Automotive & Technology engineering	SENSOR INTERFACE		=
			Ed.	manoraj.selvaraj					+ FIX
			Appr.		S220 HL HV & LV HOUSING TO WIRE HARNESS STATION-1			FMSPM00860	Page 3
Modification	Date	Name	Original		Replacement of	Replaced by			Page 32 / 32